~~**Online Resource 1**~~

Manuscript title: Integrated multi-omics and genetic analyses identify consistent direction-level mirror dysregulation between osteoarthritis and sarcopenia: development and cartilage-tier external validation of an osteoarthritis diagnostic biomarker panel

Journal: Clinical Rheumatology

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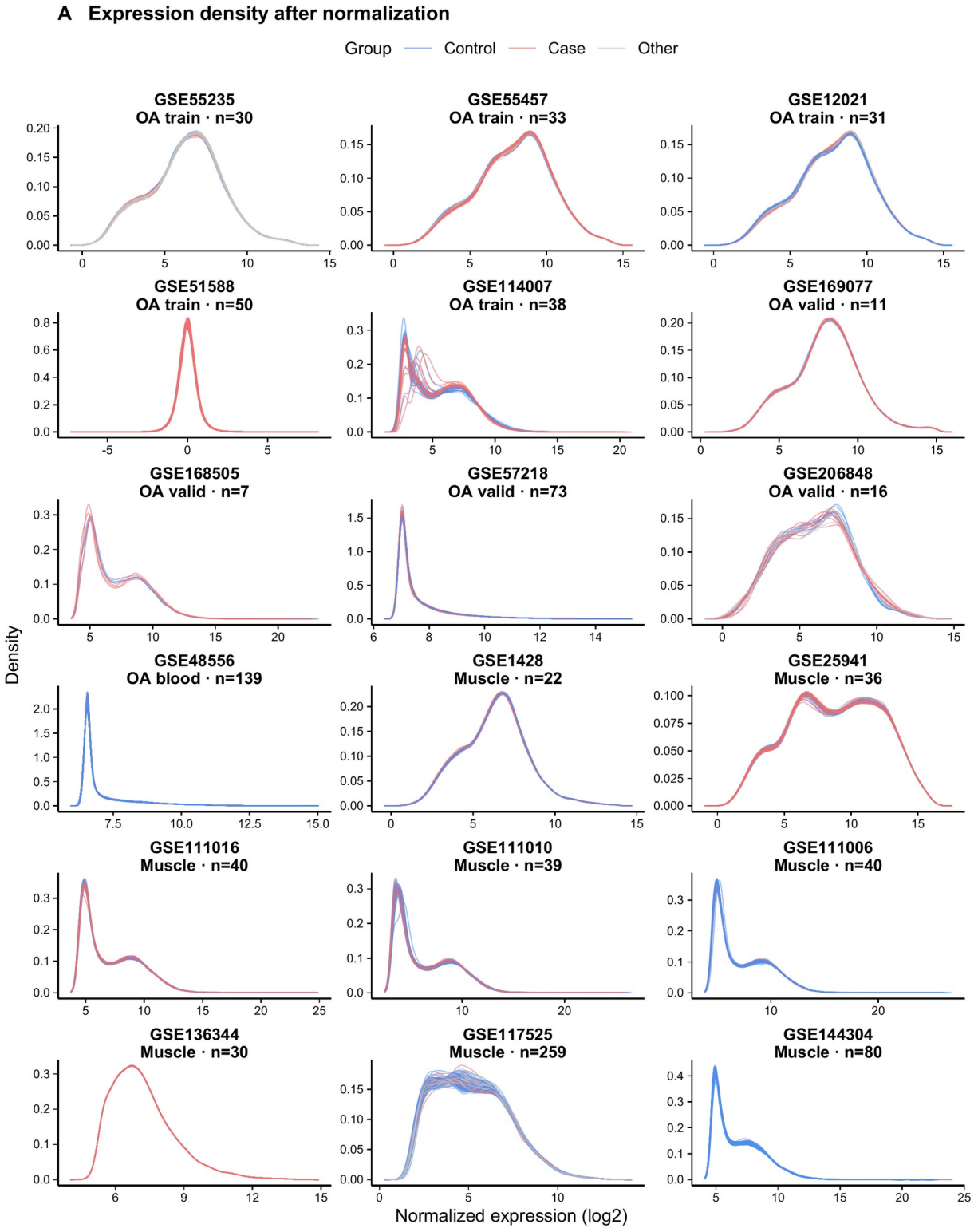
1. Supplementary figure legends (S-Fig 1–13)
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3. TRIPOD+AI (2024) reporting checklist
4. STROBE-MR (2021) checklist and Appendix D

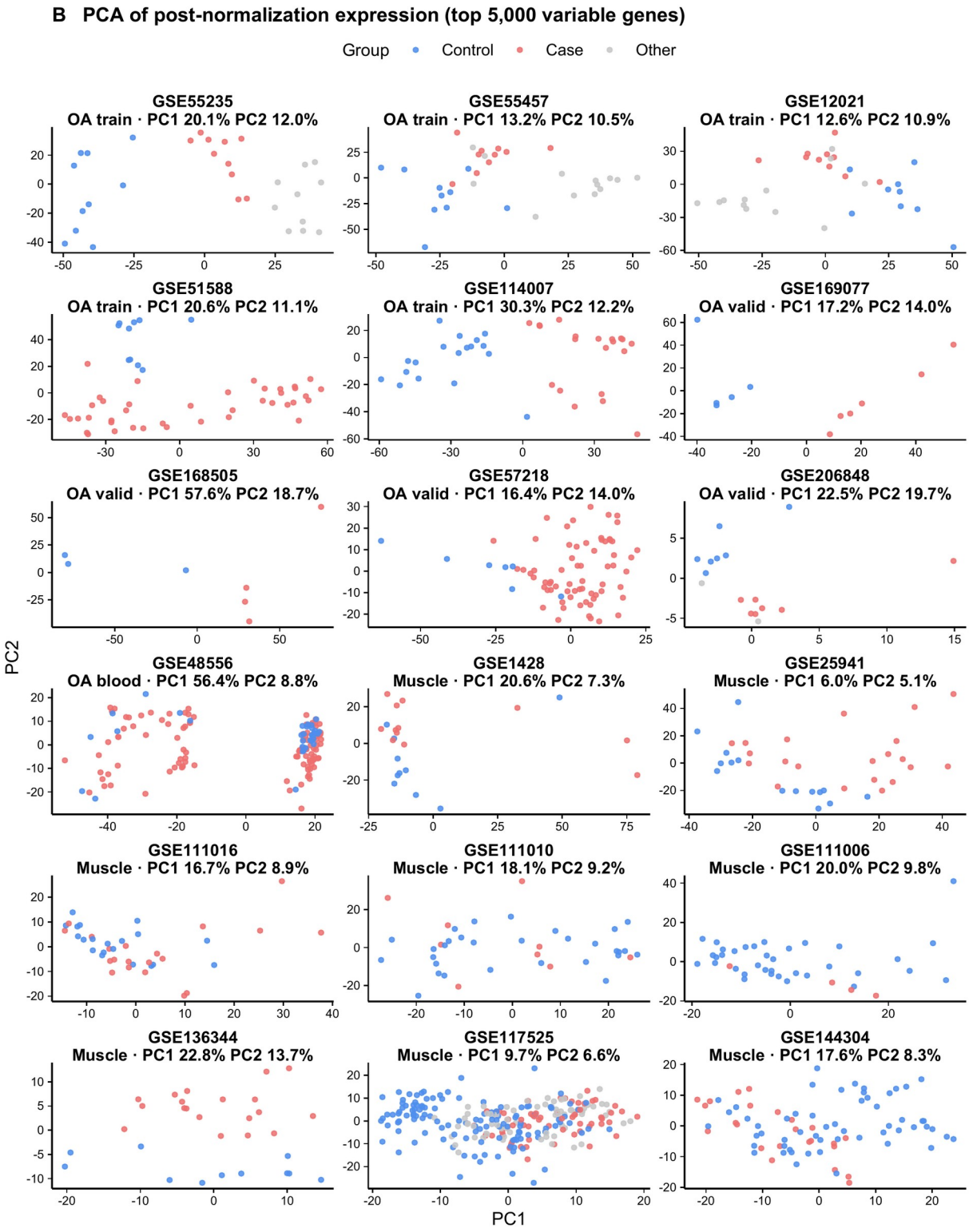
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1. Supplementary figure legends (S-Fig 1–13)

S-Fig 1 Per-dataset quality control (QC) plots after normalization for all 18 bulk transcriptome datasets. (A) Post-normalization expression density: one panel per dataset and one density curve per sample, colored by group (Control blue / Case red / Other gray); within each dataset, the sample curves were highly overlapping in shape and distribution, indicating no systematic expression shift after normalization. (B) Post-normalization principal component analysis (PCA): each panel used the top 5,000 genes by variance; points represent samples, colored by group; panel titles annotate the variance explained by PC1/PC2. The 18 datasets are ordered along four tracks: OA training (GSE55235, GSE55457, GSE12021, GSE51588,

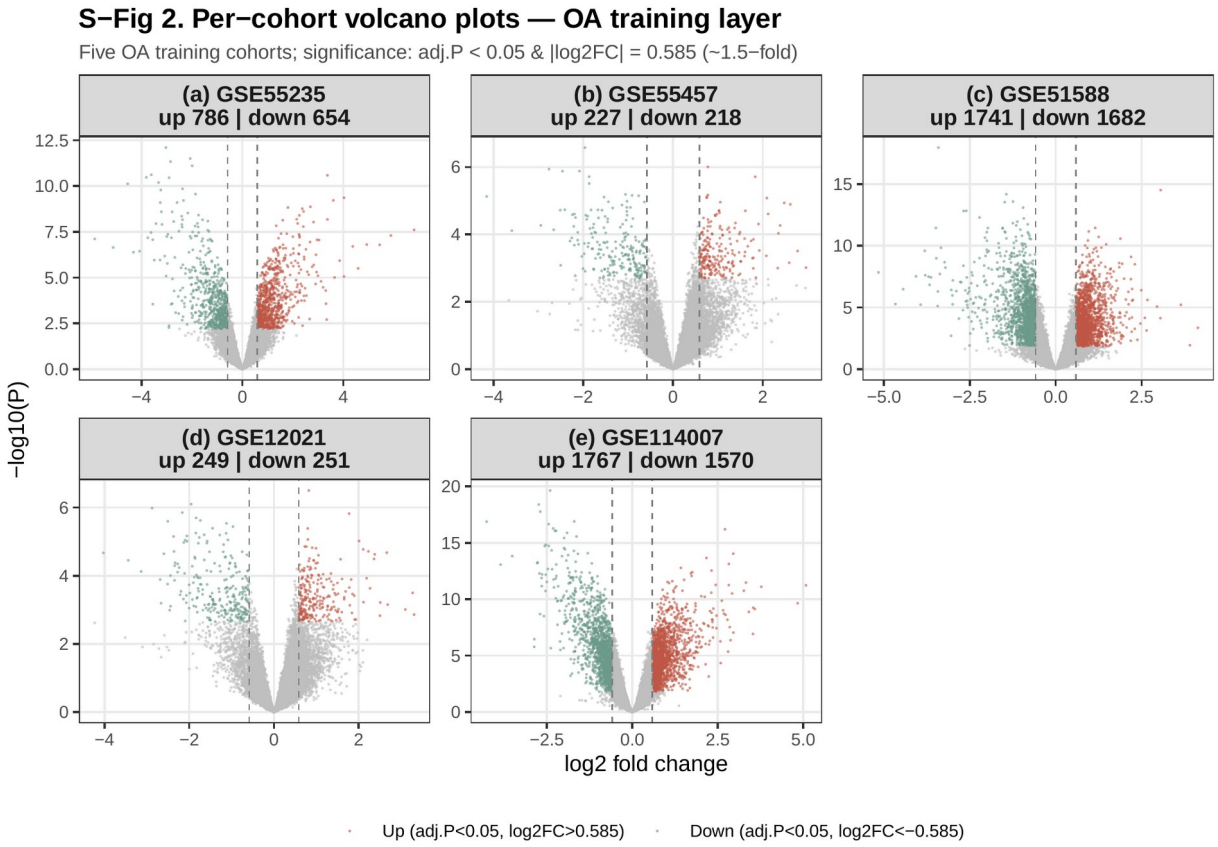
GSE114007), OA validation (GSE169077, GSE168505, GSE57218, GSE206848), OA blood (GSE48556), and skeletal muscle (GSE1428, GSE25941, GSE111016, GSE111010, GSE111006, GSE136344, GSE117525, GSE144304). This figure is a within-dataset, post-normalization QC view for inspecting normalization performance and obvious outlier samples in each dataset; the before-versus-after comparison of cross-dataset batch correction is provided by main Fig 2a (merged OA training matrix); the two are complementary and non-redundant.


A Expression density after normalization
Group Control Case Other
GSE55235
OA train · n=30
GSE55457
OA train · n=33
GSE12021
OA train · n=31
GSE51588
OA train · n=50
GSE114007
OA train · n=38
GSE169077
OA valid · n=11
GSE168505
OA valid · n=7
GSE57218
OA valid · n=73
GSE206848
OA valid · n=16
GSE48556
OA blood · n=139
GSE1428
Muscle · n=22
GSE25941
Muscle · n=36
GSE111016
Muscle · n=40
GSE111010
Muscle · n=39
GSE111006
Muscle · n=40
GSE136344
Muscle · n=30
GSE117525
Muscle · n=259
GSE144304
Muscle · n=80
Density
Normalized expression (log2)

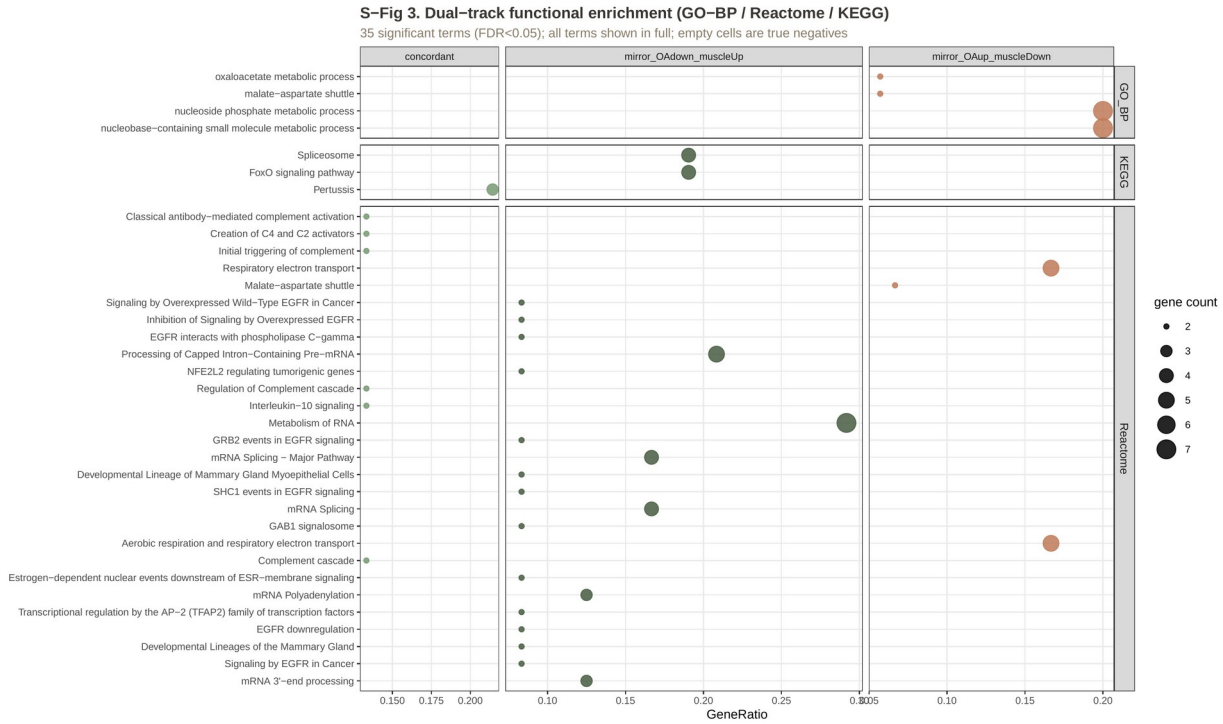


S-Fig 2 Complete overview of per-cohort differential-expression volcano plots for the OA training tier. Shown are the case–control differential-expression volcano plots for each of all 5

OA training cohorts—(a) GSE55235, (b) GSE55457, (c) GSE51588, (d) GSE12021, (e) GSE114007; main Fig 2b displays only the representative cohort GSE55235, whereas this figure provides the complete per-cohort overview. Differential-analysis software and sample-handling conventions are as in the main Methods: microarray cohorts (GSE55235/GSE55457/GSE51588/GSE12021) were analyzed with limma and the RNA-seq cohort (GSE114007) with DESeq2 for case–control differential analysis; for GSE51588, within-donor correlation of dual-site samples from the same donors was handled with duplicateCorrelation (by contrast, the biomarker training matrix retained one sample per donor to prevent donor-level leakage across cross-validation folds—the two layers' conventions differ by design); GSE12021 was restricted to the GPL96 platform. Per-cohort significant genes (adj.P<0.05 and |log2FC|≥0.585, approximately 1.5-fold) numbered 445–3,423 per cohort (descriptive statistics); points are colored by dysregulation direction and significance (upregulated/downregulated/not significant). The input to RRA cross-cohort integration consisted of per-cohort genome-wide two-sided lists ranked by P value and split by direction; integration yielded 511 integrated significant genes (adj.P<0.05), which entered the construction of the dual-track shared candidate set (leave-one-dataset-out and donor-overlap–corrected sensitivity analyses of the integrated set are given in Tables S19 and S20). The |log2FC|>1 in main Fig 2b is a display and annotation threshold for the representative cohort and does not alter the analysis inclusion conventions above. This figure is a per-cohort supplement to main Fig 2b and supports the main-text statement that “per-cohort differential analysis of the OA training tier yielded 445–3,423 significant genes per cohort.”

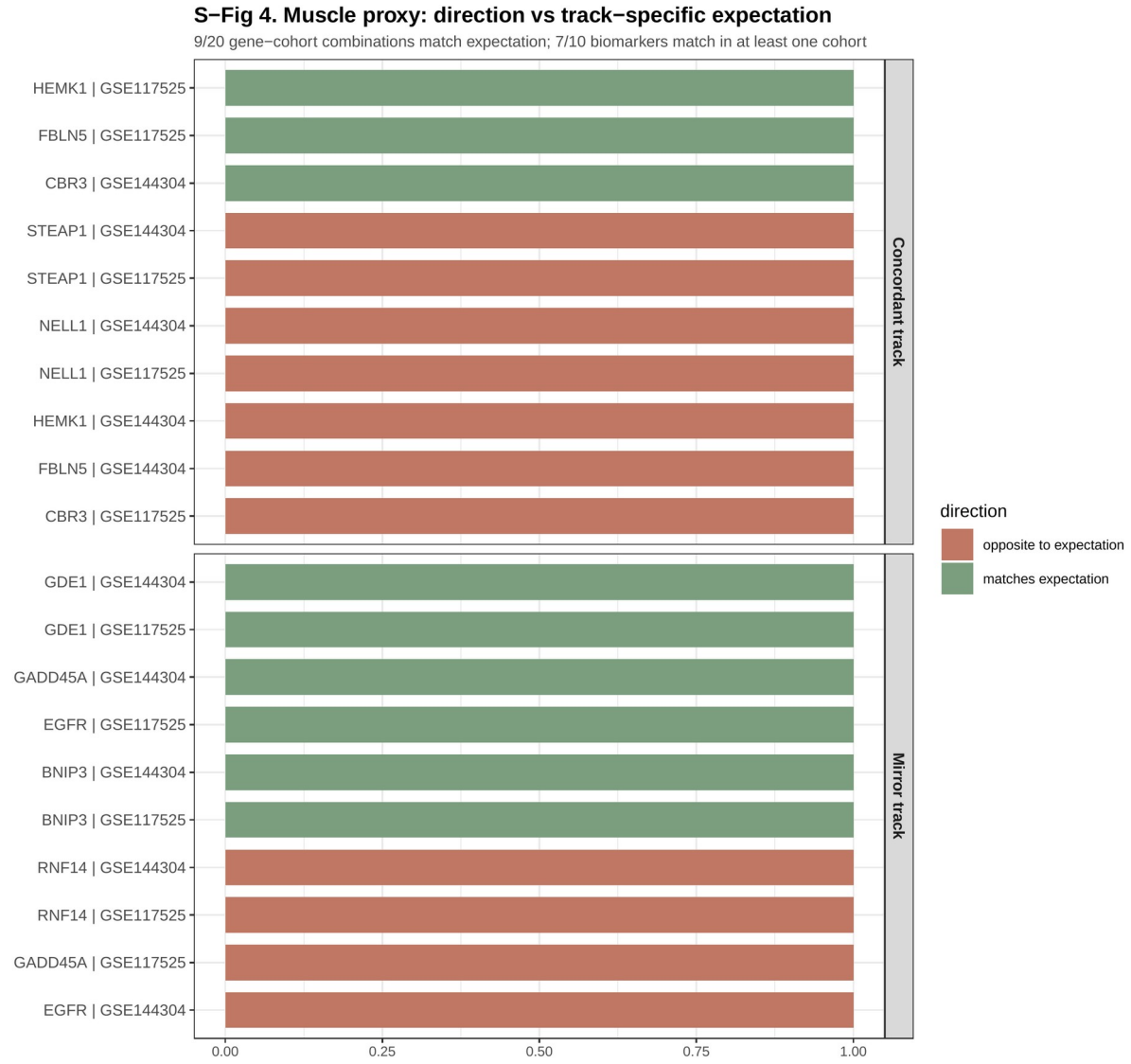


S-Fig 3 Dot-plot overview of all significant terms from track-stratified functional enrichment. Shown are all 35 FDR<0.05 significant terms (GO-BP 4, Reactome 28, KEGG 3) from over-representation analysis (ORA, clusterProfiler) across three tracks (concordant track, OA-up/muscle-down mirror track, OA-down/muscle-up mirror track) × three databases (GO-BP, Reactome, KEGG), arranged in full by track × database facets; this is a single-page overview of the same significant set as main Fig 2e/2f. Analysis conventions are consistent with the main Methods: the background gene set comprised all genes involved in dual-track construction (9,960), with BH correction and threshold FDR<0.05; KEGG used a local snapshot (2026-08-23, 372 pathways / 9,423 Entrez genes); the absence of significant KEGG terms for the OA-up/muscle-down mirror track is a true negative. Full per-term statistics including non-significant terms are provided in the repository-hosted intermediate tables `results/moduleC/compareCluster_*_full.csv` (the significant-term summary is `enrich_ALL_tracks_significant.csv`).



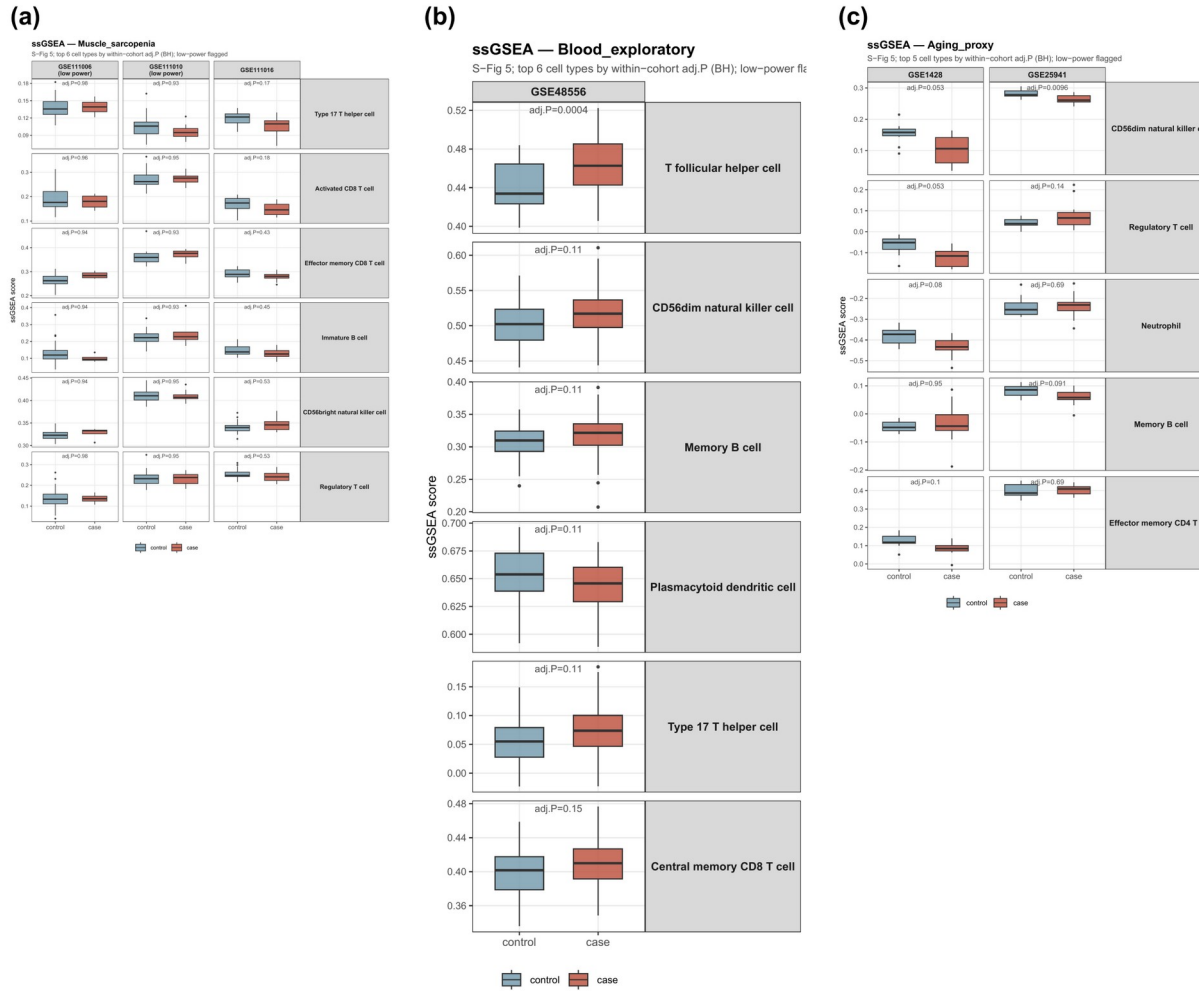
S-Fig 4 Mirror-direction concordance comparison of the 10 biomarkers in the muscle-proxy exploratory tier. Shown is the comparison of the expression directions of the 10 biomarkers (NELL1, STEAP1, GADD45A, FBLN5, HEMK1, GDE1, RNF14, BNIP3, EGFR, CBR3) in muscle-proxy cohorts (frailty-proxy GSE117525/GSE144304; aging-proxy GSE136344) against the directions expected from the discovery-tier dual tracks (including mirror-track expectations). Per the pre-specified tiered validation framework, the muscle-proxy exploratory tier had no performance threshold and served for direction and lead reporting only; directional concordance was accepted when at least 2 cohorts agreed. The overall directional concordance rate was 9/20 (45%), numerically below the 50% chance expectation (two-sided binomial P = 0.82); under the prespecified acceptance criterion (directional concordance in ≥ 2 cohorts) only 2/10 biomarkers (GDE1 and BNIP3, concordant in all available cohorts) were accepted, so this tier provided no supportive evidence for the panel; for completeness, when no cohort-count requirement was applied, 7/10 biomarkers matched track expectations in at least one proxy cohort; NELL1, STEAP1, and RNF14 received no directional support; the concordant signals were concentrated in the OA-down/muscle-up mirror track (4/6, 66.7%), a descriptive pattern compatible with the direction of the FoxO atrophy axis but not constituting evidence. GSE136344 had an empty intersection with the biomarkers (0/0) and contributed no directional statistics. This tier

constitutes frailty/aging proxy-phenotype evidence and does not constitute validation at the level of confirmed sarcopenia; performance claims of AUC greater than 0.7 apply only to the cartilage primary validation layer.

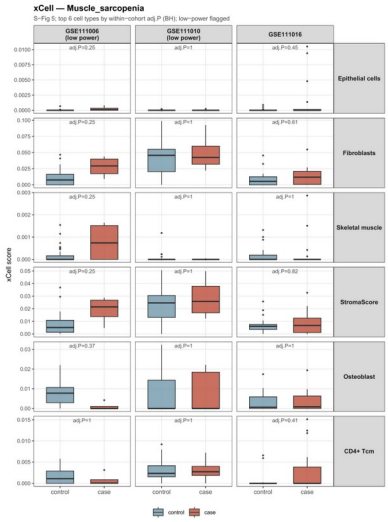


S-Fig 5 Group-comparison boxplot panels for the immune-infiltration exploratory tiers (three methods × three exploratory tiers, 9 panels in total). Shown are per-sample cell-score group-comparison boxplots for three algorithms—ssGSEA (a–c), xCell (d–f), and CIBERSORT (g–i)—across three exploratory tiers: the muscle tier (Muscle_sarcopenia: GSE111006/GSE111010/GSE111016), the blood tier (Blood_exploratory: GSE48556), and the aging proxy tier (Aging_proxy: GSE1428/GSE25941); each tier displays the cell types ranking highest by ascending adjusted P value from within-cohort group comparisons. Statistical conventions are consistent with the main Methods: group comparisons were performed within cohorts only (limma moderated t); BH correction was applied within each cohort × method; low-power cohorts (GSE111006, GSE111010; fewer than 10 cases) are labeled low_power;

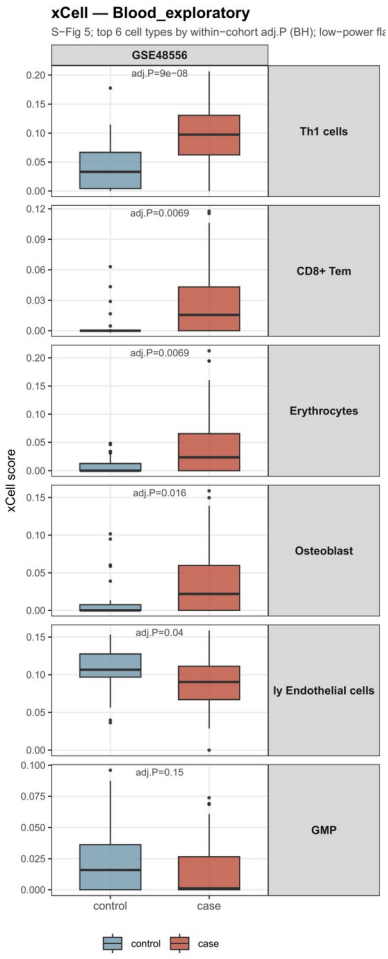
CIBERSORT was the open-source ν-SVR implementation (dicepro v1.0.2; LM22 signature matrix: 547 marker genes, 22 immune cell subsets; 100 permutations; deconvolution input was linear-scale expression values). Key cell-type results for the high-confidence tiers (the OA tissue tier and the QC full-pass blood tier) are presented in the main text and S-Fig 11; this figure presents the exploratory tiers in full: in the cross-cohort directional summary (468 entries), no reproducible signal was detected in the muscle, blood, or aging tiers; blood-tier CIBERSORT sample QC passed in full (139/139) yet group comparisons were entirely non-significant, consistent with the tissue-gap interpretation. The overall CIBERSORT sample QC ($P<0.05$) pass rate was 45.1% (463 samples); signals in the RNA-seq tier (including GSE114007) were labeled "QC-filtered, exploratory only" per QC criteria and entered the supplementary material only, not the main conclusions. Muscle-tier results are stated as "not detected at the current sample size"; aging proxy tier results must not be written as sarcopenia evidence; all results in this section are exploratory.


(a)
ssGSEA — Muscle_sarcopenia
S−Fig 5; top 6 cell types by within−cohort adj.P (BH); low−power flagged
GSE111006 (low power!)
GSE111010 (low power!)
GSE111016
Type 17 T helper cell
Activated CD8 T cell
Effector memory CD8 T cell
Immature B cell
CD56bright natural killer cell
Regulatory T cell
ssGSEA score
control
case
(b)
ssGSEA — Blood_exploratory
S−Fig 5; top 6 cell types by within−cohort adj.P (BH); low−power fla
GSE48556
adj.P=0.0004
T follicular helper cell
adj.P=0.11
CD56dim natural killer cell
adj.P=0.11
Memory B cell
adj.P=0.11
Plasmacytoid dendritic cell
adj.P=0.11
Type 17 T helper cell
adj.P=0.15
Central memory CD8 T cell
ssGSEA score
control
case
(c)
ssGSEA — Aging_proxy
S−Fig 5; top 5 cell types by within−cohort adj.P (BH); low−power flagged
GSE1428
GSE25941
adj.P=0.053
adj.P=0.0096
CD56dim natural killer
adj.P=0.053
adj.P=0.14
Regulatory T cell
adj.P=0.08
adj.P=0.69
Neutrophil
adj.P=0.95
adj.P=0.091
Memory B cell
adj.P=0.1
adj.P=0.69
Effector memory CD4 T
ssGSEA score
control
case

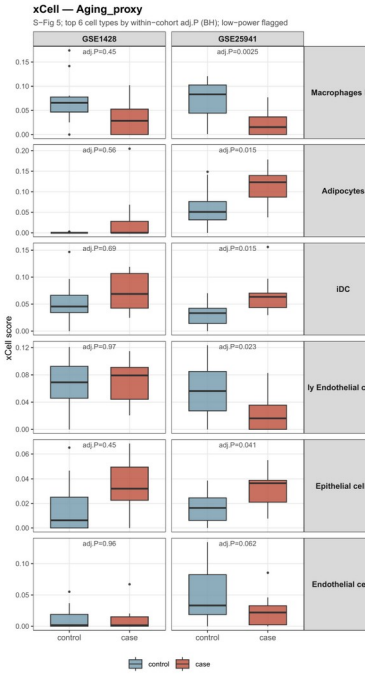
(d)

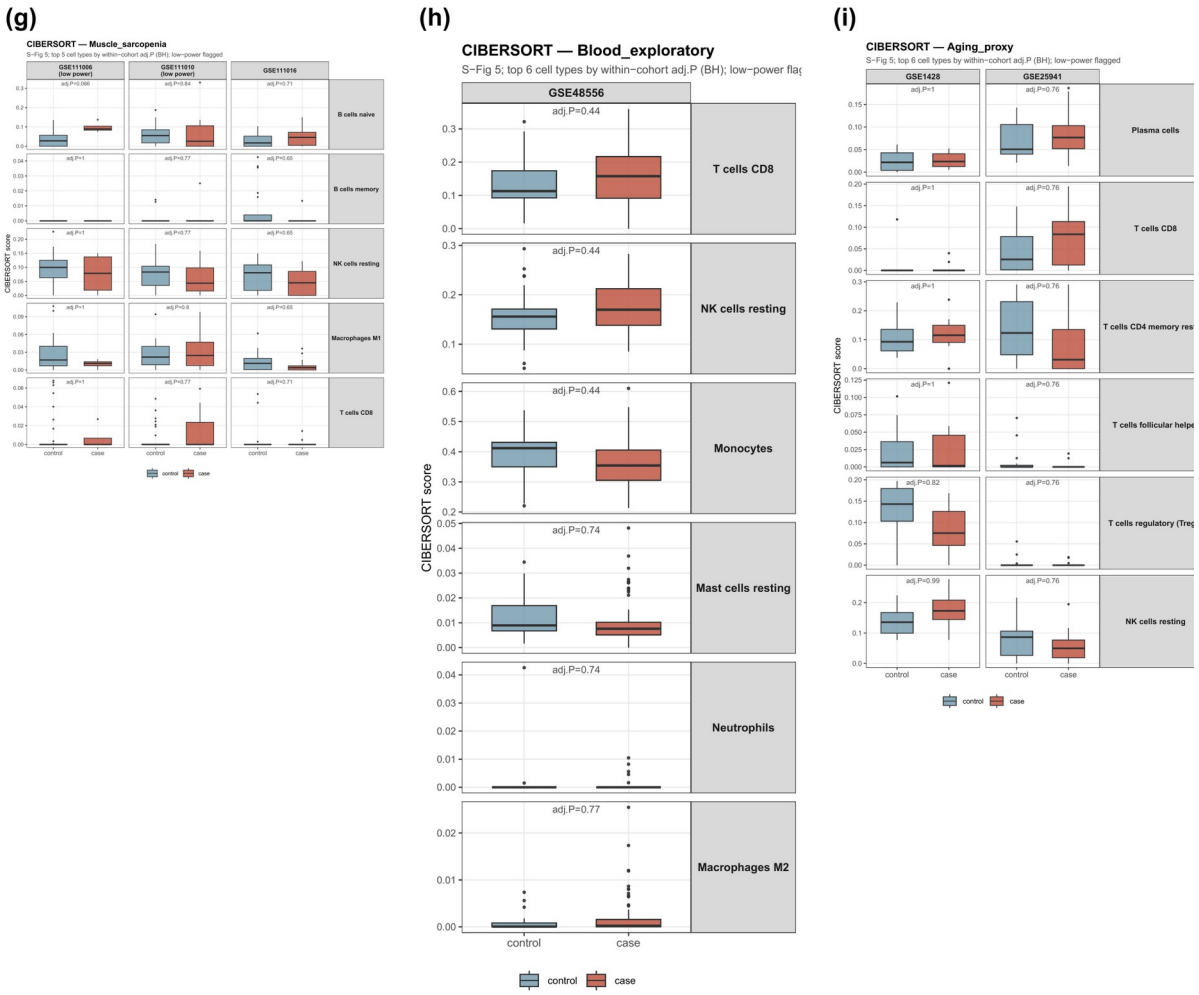

xCell — Muscle_sarcopenia
S−Fig 5; top 6 cell types by within−cohort adj.P (BH); low−power flagged
GSE111006 (low power)
GSE111010 (low power)
GSE111016
Epithelial cells
Fibroblasts
Skeletal muscle
StromaScore
Osteoblast
CD4+ Tcm
xCell score
control
case

(e)


xCell — Blood_exploratory
S−Fig 5; top 6 cell types by within−cohort adj.P (BH); low−power fla
GSE48556
adj.P=9e−08
Th1 cells
adj.P=0.0069
CD8+ Tem
adj.P=0.0069
Erythrocytes
adj.P=0.016
Osteoblast
adj.P=0.04
ly Endothelial cells
adj.P=0.15
GMP
xCell score
control
case

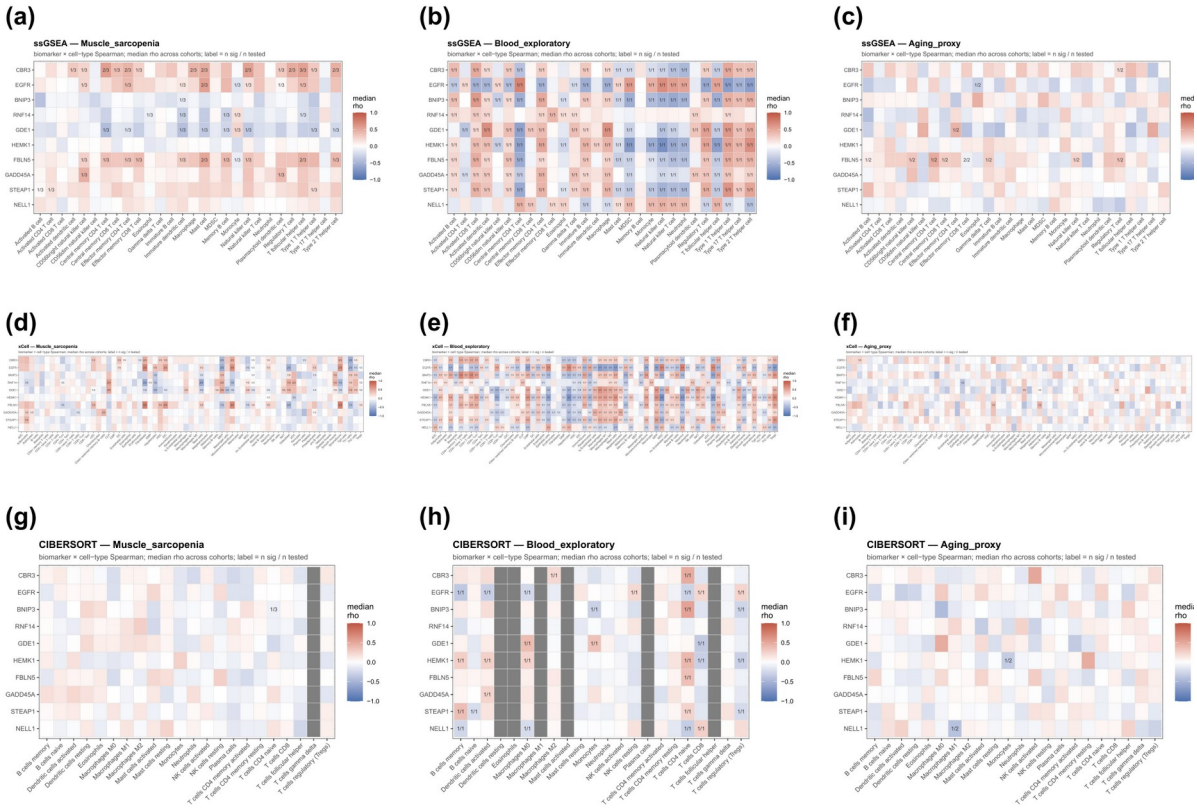
(f)


xCell — Aging_proxy
S−Fig 5; top 6 cell types by within−cohort adj.P (BH); low−power flagged
GSE1428
GSE25941
Macrophages
Adipocytes
iDC
Epithelial cel
Endothelial ce
xCell score
control
case



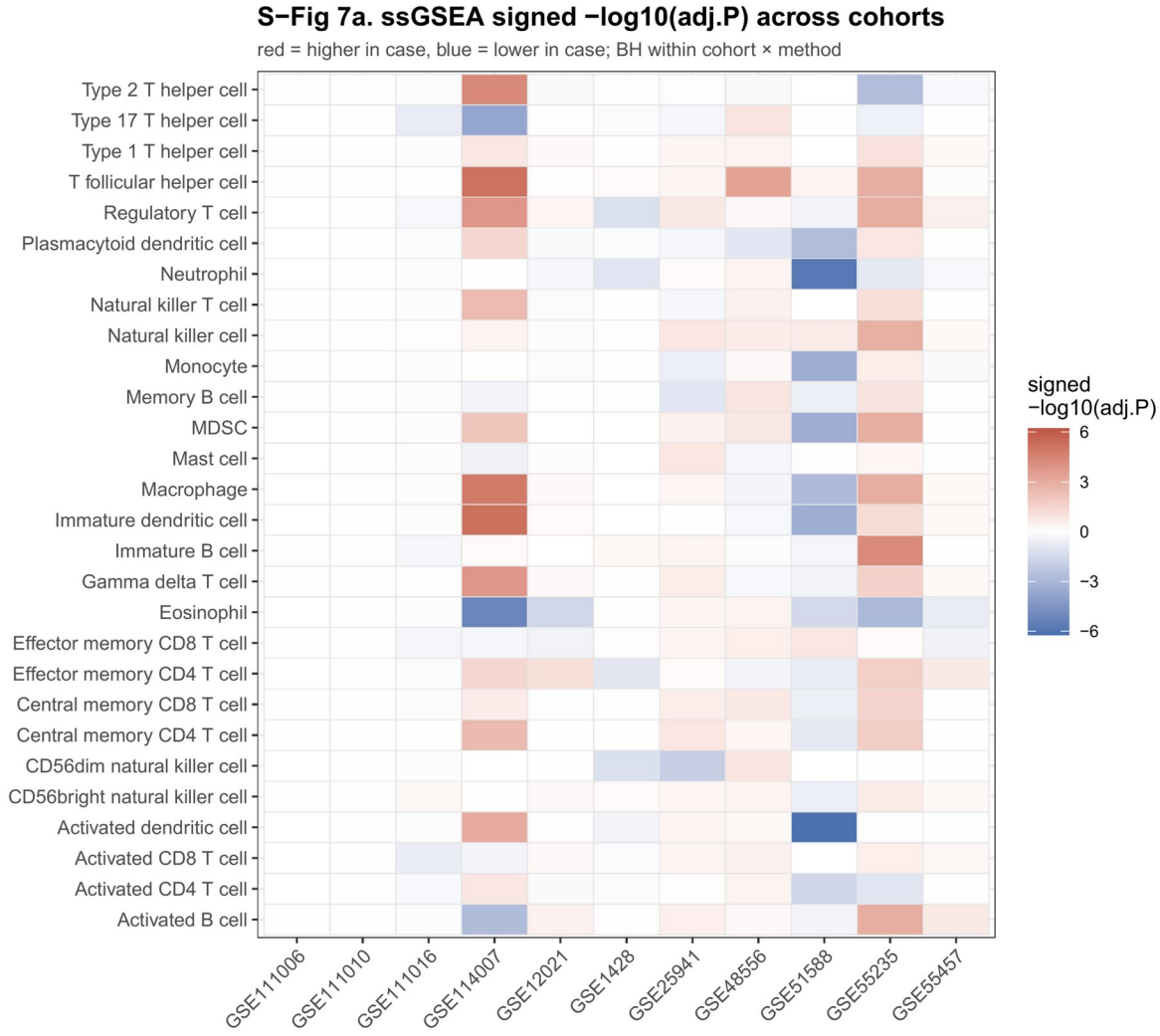
S-Fig 6 Biomarker × cell-type correlation heatmap panels for the immune-infiltration exploratory tiers (three methods × three exploratory tiers, 9 panels in total). Shown are within-cohort Spearman correlation heatmaps of the expression of the 10 biomarkers against the scores of individual cell types in the muscle, blood, and aging proxy tiers, computed by the three algorithms ssGSEA (a–c) / xCell (d–f) / CIBERSORT (g–i); correlation tests were BH-corrected within each cohort × method; combinations with insufficient computable sample size (n<8) or zero variance were recorded as missing, with no uninformative inference drawn. The corresponding correlation heatmap for the OA training tier is shown in S-Fig 11c; the biomarker–cell correlation signals reported in the main text (GDE1/RNF14 correlated with the T cell–macrophage axis, concordant in 5/5 cohorts, 3/5 passing FDR; FBLN5/EGFR correlated with fibroblasts/StromaScore, significant in 3/3; CBR3 correlated with Tfh, significant in 3/3) all

derive from the OA tissue tier. Exploratory-tier correlations are reported as leads only, with no reproducibility claims; aging proxy tier results must not be written as sarcopenia evidence.

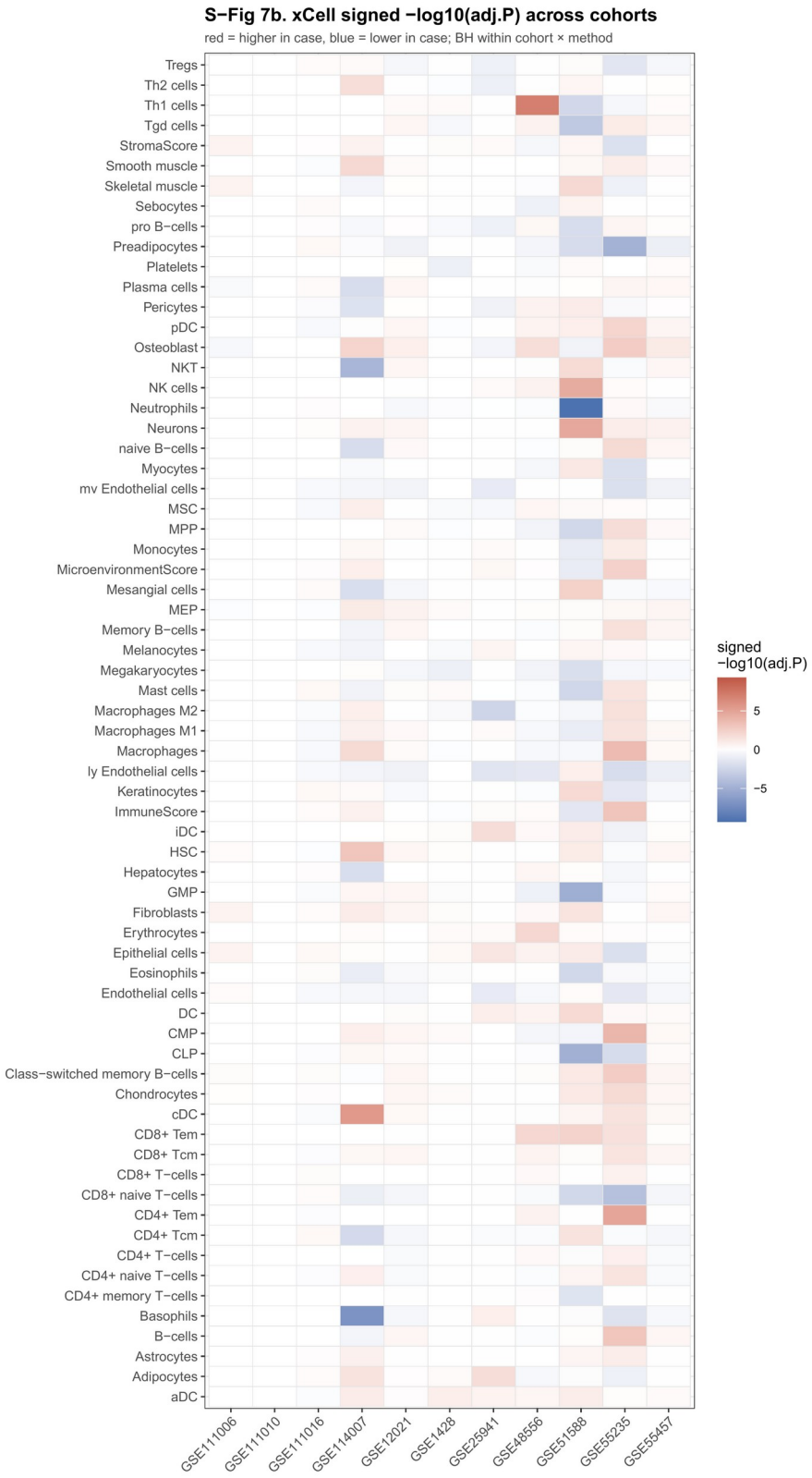


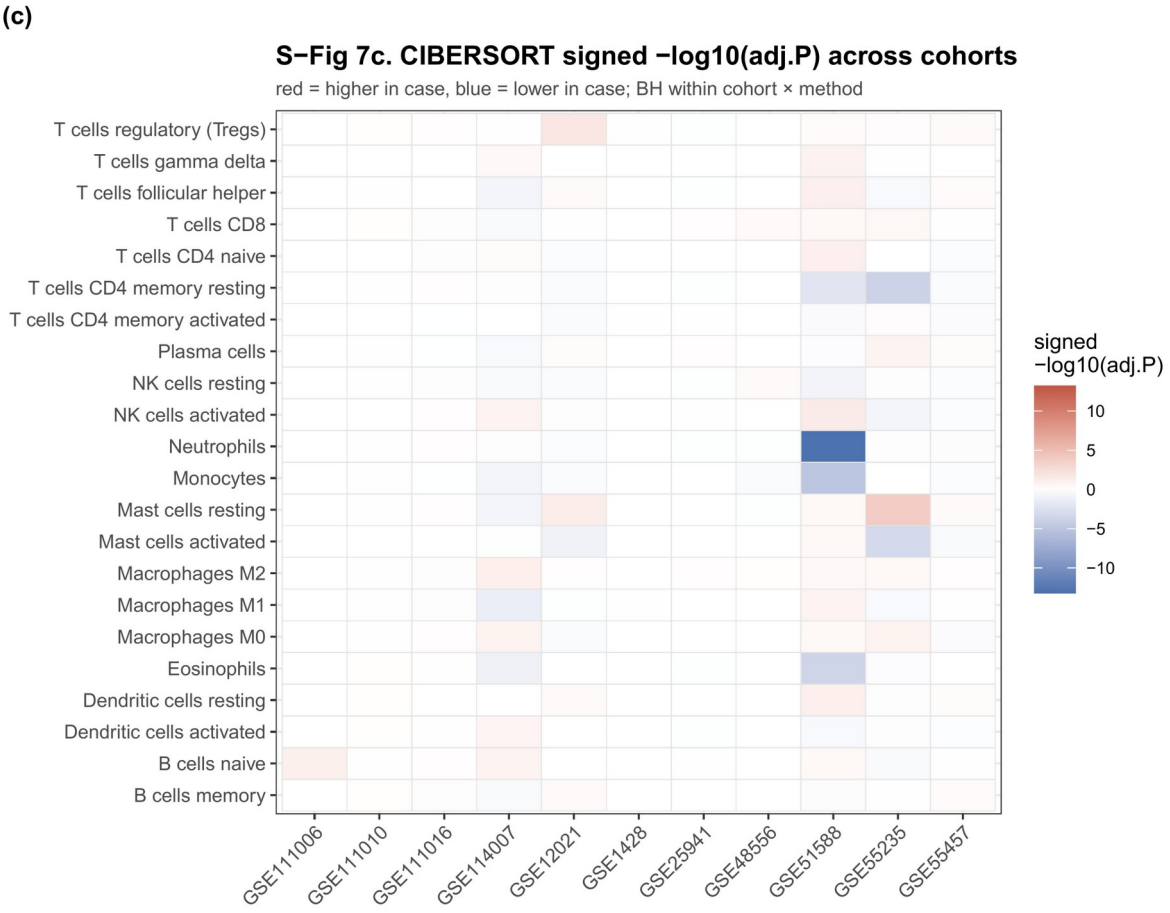
S-Fig 7 Cross-cohort directional-concordance signed heatmaps (per-method view). Shown are cohort × cell-type signed heatmaps for each of the three algorithms—(a) ssGSEA, (b) xCell, (c) CIBERSORT: cell values are the direction-weighted significance of group differences (sign(Δ) × −log10(adjP), positive values = elevated in the case group), BH-corrected within each cohort × method, with missing combinations shown in gray. This figure is the per-method base view of the cross-cohort directional summary of 468 entries (tier × method × cell_type): per the pre-specified grading rules, entries with at least 2 cohorts significant in the same direction and none significant in the opposite direction were graded replicated, those significant in both directions were graded conflicting, and the remainder were graded exploratory/none; cross-cohort summarization pooled directions only, without pooled P. All 16 replicated signals were located in the OA tissue tier; no reproducible signal was detected in the muscle, blood, or aging tiers; the verdict summary heatmap for three methods × four tissue tiers is shown in S-Fig 11a.

(a)


S−Fig 7a. ssGSEA signed −log10(adj.P) across cohorts
red = higher in case, blue = lower in case; BH within cohort × method
Type 2 T helper cell
Type 17 T helper cell
Type 1 T helper cell
T follicular helper cell
Regulatory T cell
Plasmacytoid dendritic cell
Neutrophil
Natural killer T cell
Natural killer cell
Monocyte
Memory B cell
MDSC
Mast cell
Macrophage
Immature dendritic cell
Immature B cell
Gamma delta T cell
Eosinophil
Effector memory CD8 T cell
Effector memory CD4 T cell
Central memory CD8 T cell
Central memory CD4 T cell
CD56dim natural killer cell
CD56bright natural killer cell
Activated dendritic cell
Activated CD8 T cell
Activated CD4 T cell
Activated B cell
GSE111006
GSE111010
GSE111016
GSE114007
GSE12021
GSE1428
GSE25941
GSE48556
GSE51588
GSE55235
GSE55457
signed
−log10(adj.P)
6
3
0
−3
−6

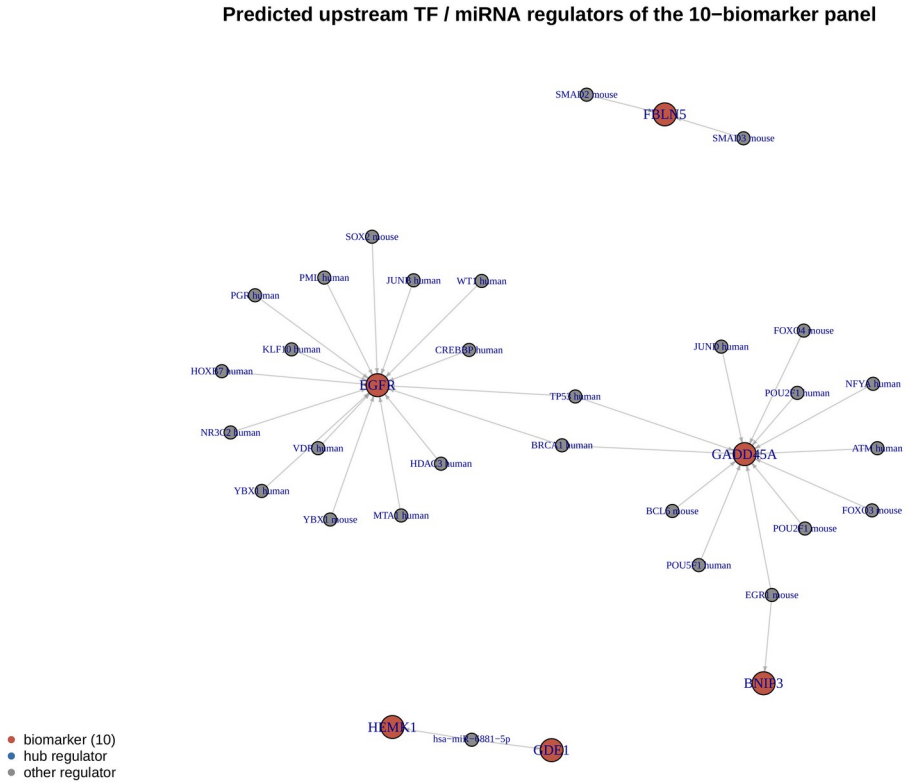
(b)


S−Fig 7b. xCell signed −log10(adj.P) across cohorts
red = higher in case, blue = lower in case; BH within cohort × method
Tregs
Th2 cells
Th1 cells
Tgd cells
StromaScore
Smooth muscle
Skeletal muscle
Sebocytes
pro B−cells
Preadipocytes
Platelets
Plasma cells
Pericytes
pDC
Osteoblast
NKT
NK cells
Neutrophils
Neurons
naive B−cells
Myocytes
mv Endothelial cells
MSC
MPP
Monocytes
MicroenvironmentScore
Mesangial cells
MEP
Memory B−cells
Melanocytes
Megakaryocytes
Mast cells
Macrophages M2
Macrophages M1
Macrophages
ly Endothelial cells
Keratinocytes
ImmuneScore
iDC
HSC
Hepatocytes
GMP
Fibroblasts
Erythrocytes
Epithelial cells
Eosinophils
Endothelial cells
DC
CMP
CLP
Class−switched memory B−cells
Chondrocytes
cDC
CD8+ Tem
CD8+ Tcm
CD8+ T−cells
CD8+ naive T−cells
CD4+ Tem
CD4+ Tcm
CD4+ T−cells
CD4+ naive T−cells
CD4+ memory T−cells
Basophils
B−cells
Astrocytes
Adipocytes
aDC
GSE111006
GSE111010
GSE111016
GSE114007
GSE12021
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GSE55235
GSE55457
signed −log10(adj.P)
5
0
−5



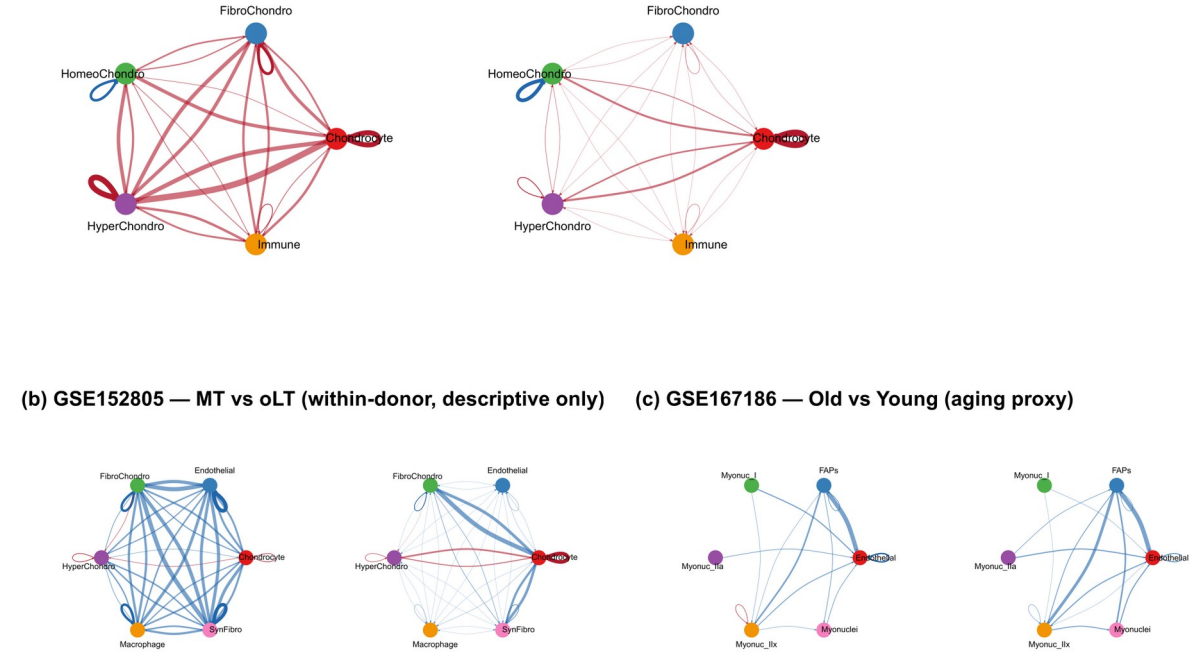
S-Fig 8 Upstream TF–miRNA regulatory network of the 10 biomarkers. TRRUST_Transcription_Factors_2019 and ChEA_2022 (dual TF databases) and miRTarBase_2017 (experimentally validated miRNA–target genes) were queried via the Enrichr interface, with significance threshold adj.P<0.05; for databases not reaching the threshold, the top 10 nominally significant entries were taken and flagged nominal_only in the edge table; TFs jointly indicated by multiple sources were marked as a high-confidence tier. In total, 75 regulatory edges were obtained (73 TF, 2 miRNA), covering 6/10 biomarkers; NELL1, STEAP1, RNF14, and CBR3 had no recorded significant upstream regulators. No regulator targeted ≥3/10 biomarkers, suggesting the absence of a single master regulator, compatible with the multi-axis parallel mirror pattern. In the figure, nodes represent regulators and biomarkers, and edges represent database-recorded regulatory relationships; for readability, the network plots only adj.P<0.05 significant edges, and nominal_only edges are not shown; the complete edge table (with nominal_only flags) is provided in Table S14; the network layout was based on igraph with the random seed explicitly fixed. This analysis is a database-driven predictive result, positioned

as hypothesis-generating for upstream regulation, with no mechanistic assertions; the complete edge table and hub summary are provided in Table S14.



S-Fig 9 Differential cell–cell communication networks for three single-cell/single-nucleus datasets (CellChat v2.2, full CellChatDB.human database, triMean approximation, population.size=TRUE, filterCommunication min.cells=10, random seed explicitly fixed). (a) GSE169454 chondrocyte scRNA-seq primary analysis tier: comparison of the OA group versus normal controls showed 505 vs 69 ligand–receptor interactions, indicating markedly increased communication in OA cartilage; no significant interactions involving the 10 biomarkers as ligands or receptors were detected in the primary analysis tier, indicating that the communication-tier remodeling is not directly mediated by this panel. (b) GSE152805 descriptive tier: within-donor comparison only, with 1761 interactions in the MT region and 282 in the oLT region; no healthy controls were available, and only communication-pattern differences are described. (c) GSE167186 exploratory tier (aging proxy): comparison of Old versus Young detected an EGF→EGFR interaction (type IIa/IIx myonuclei → FAPs, present in both groups), forming an exploratory echo with the compartment preference of EGFR in muscle FAPs; this evidence is constrained by the aging-proxy positioning and supports no sarcopenia

conclusion. Both-group comparisons describe communication-pattern differences only, with no causal assertions; all results in this section are exploratory. The full table of biomarker-related L–R interactions and the per-dataset group-comparison summary are provided in Table S15.

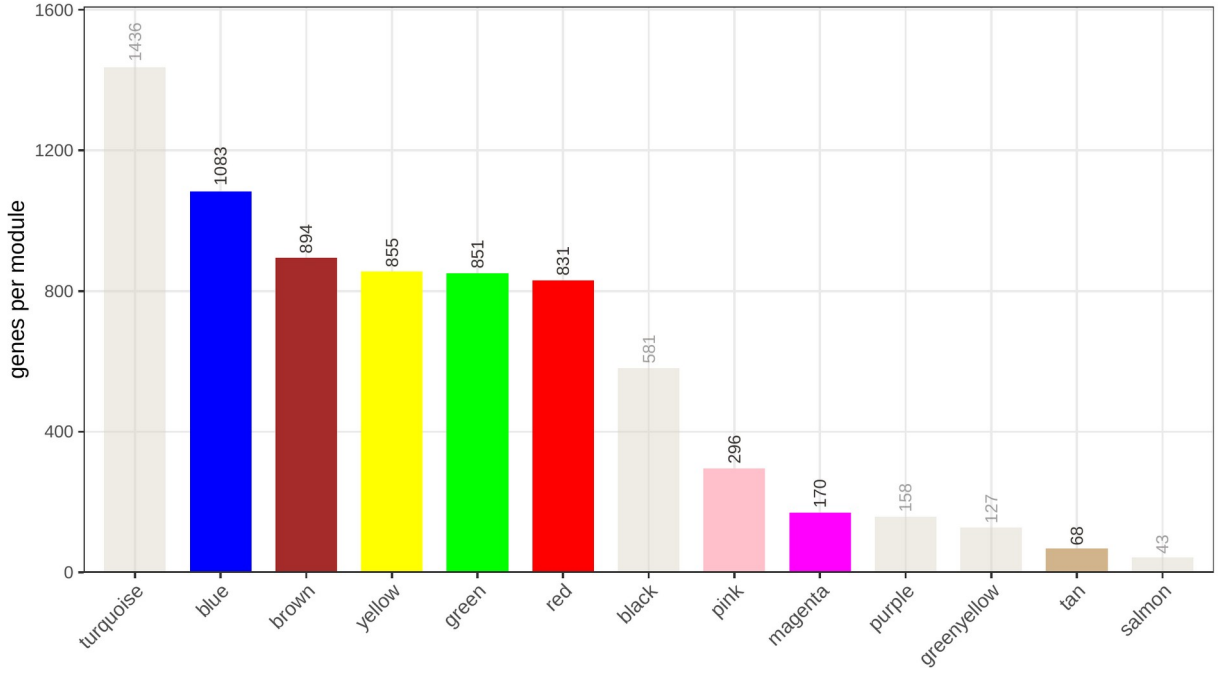


Note: red = stronger, blue = weaker (2nd group vs 1st, CellChat differential network); edge width = strength.

S-Fig 10 WGCNA weighted gene co-expression network and module localization of dual-track genes (confidence-annotation framing). (a) OA-side module-size bar plot: gene counts of non-grey modules, with disease-associated key modules shown in solid color (8 key modules); (b) muscle-side (GSE111016) module-size bar plot (4 key modules); (c) bubble plot of module membership of dual-track candidate genes: assignment counts of concordant-track and mirror-track genes to individual modules on the OA side and the muscle side (excluding the grey module). Soft-threshold powers were selected by the scale-free topology fit criterion $R^2>0.8$ (muscle side GSE111016 used soft power=8). The OA side yielded 8 disease-associated modules (2,119 high-confidence genes, $|kME|\geq0.7$ and $GS\geq0.3$); the muscle side yielded 4 modules (304 high-confidence genes); among dual-track genes, 7/27 concordant-track and 27/72 mirror-track genes fell within high-confidence modules. This tier is positioned as confidence annotation rather than hard-intersection filtering and does not alter dual-track membership.

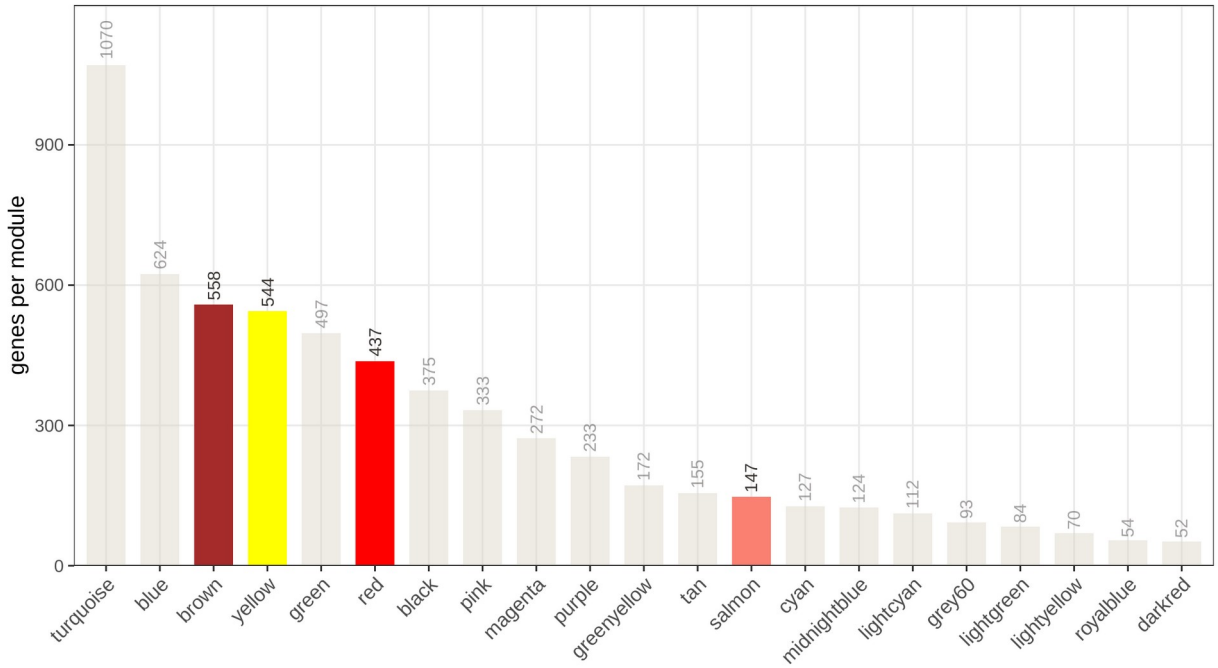
S−Fig 10a. WGCNA modules — OA (confidence annotation layer)

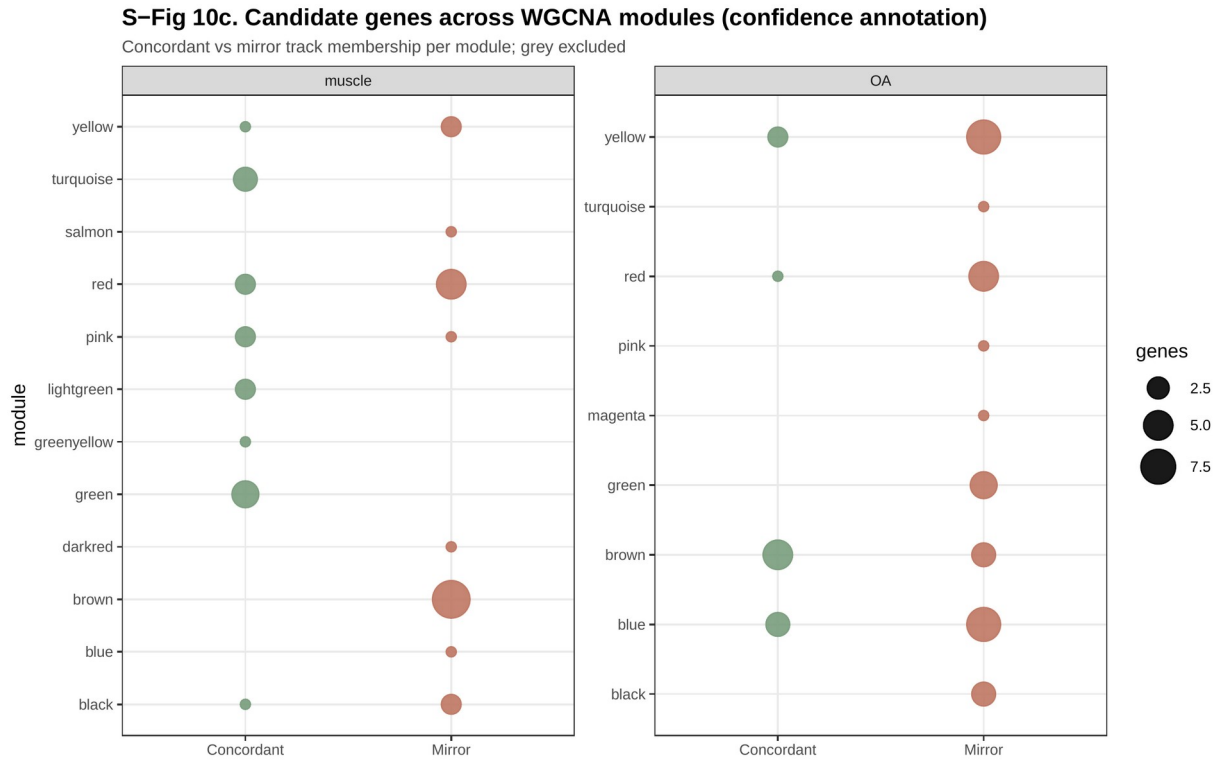
8 key modules (solid) with 2119 high−confidence genes (|kME|>=0.7 & GS>=0.3); 13 non−grey modules total



S−Fig 10b. WGCNA modules — muscle (confidence annotation layer)

4 key modules (solid) with 304 high−confidence genes (|kME|>=0.7 & GS>=0.3); 21 non−grey modules total





S-Fig 11 Cross-method, cross-cohort summary of immune infiltration and display of key cell types. (a) Cross-cohort directional-concordance heatmap: verdict summary of three algorithms (ssGSEA, xCell, and the open-source ν-SVR CIBERSORT implementation) × four tissue tiers (OA tissue, muscle, blood, aging) (the cross-cohort directional summary comprised 468 entries, graded replicated/conflicting/exploratory); all 16 replicated signals were located in the OA tissue tier, and no reproducible signal was detected in the muscle, blood, or aging tiers; (b) group-comparison boxplots of key cell types: per-sample scores from the high-confidence tiers (blood 139/139 and three synovial cohorts 59/59 samples passing QC), independently re-run and validated on the official Stanford CIBERSORTx platform (official LM22 signature file, relative mode, 100 permutations), with a directional concordance rate of 92.0% across 88 cohort × cell-type combinations; (c) biomarker × cell-type Spearman correlation heatmap (OA training tier), with correlation tests BH-corrected. The 2 CIBERSORT replicated signals (elevated resting mast cells, decreased resting memory CD4 T cells) were independently replicated by this method only; xCell and ssGSEA did not replicate in the same direction, so no three-method consensus statement is made.

S−Fig 11a (redraw). Cross−cohort direction−consistency verdicts

97 non−none entries shown (of 468); replicated signals (16) all in OA_training; label = n sig up / n sig down

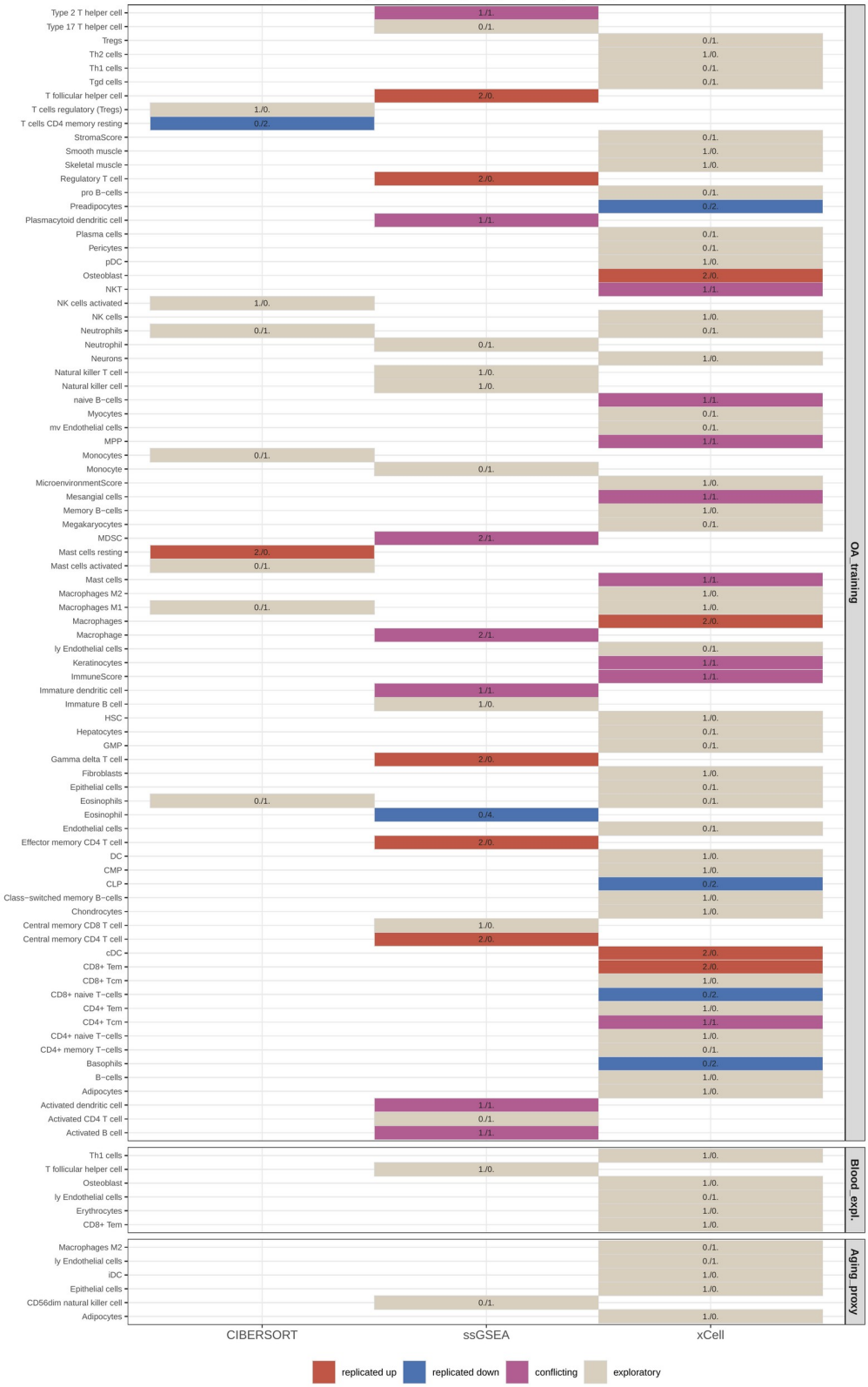


Fig 7b. Key cell−type fractions (official CIBERSORTx, per−sample)

case vs control; key replicated cell types across 4 validation cohorts

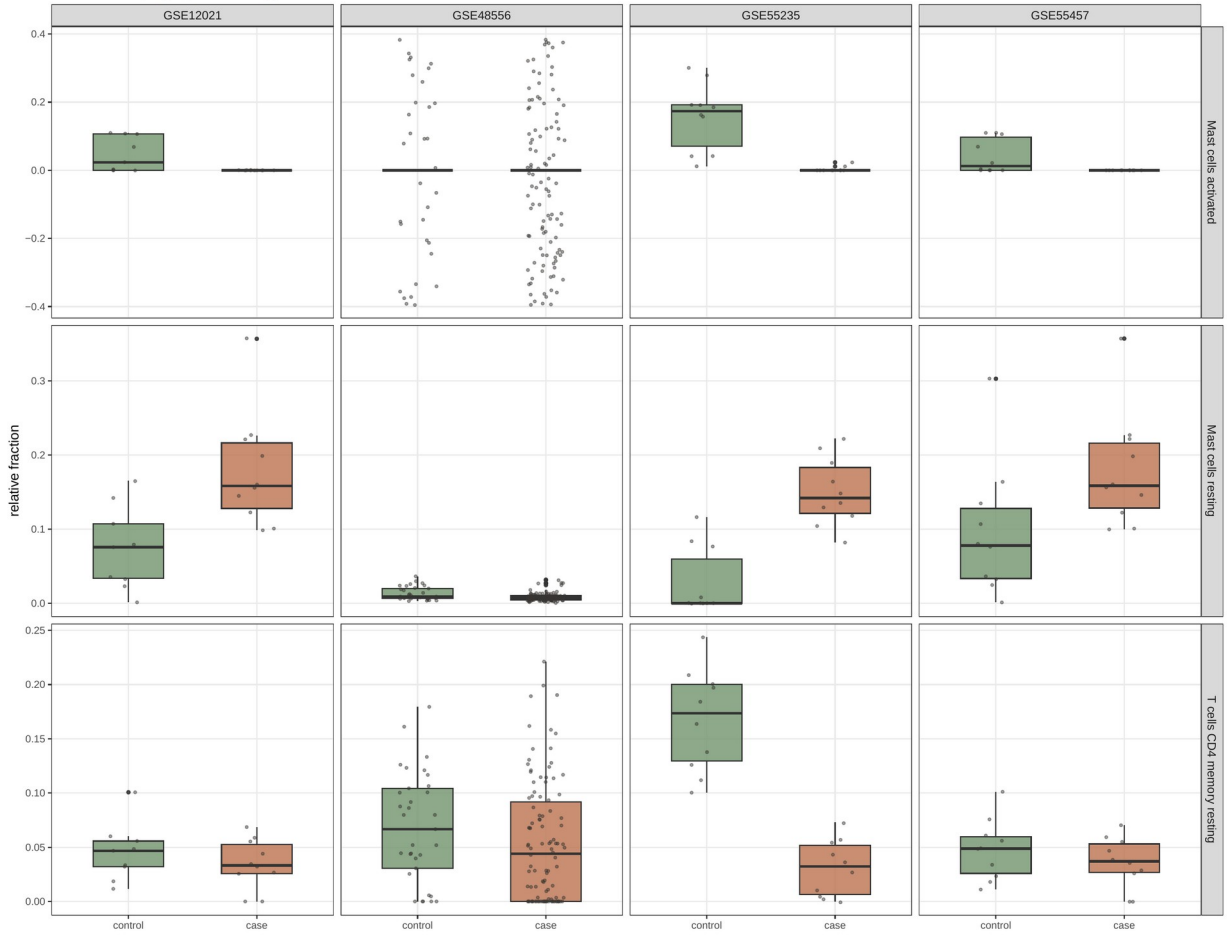
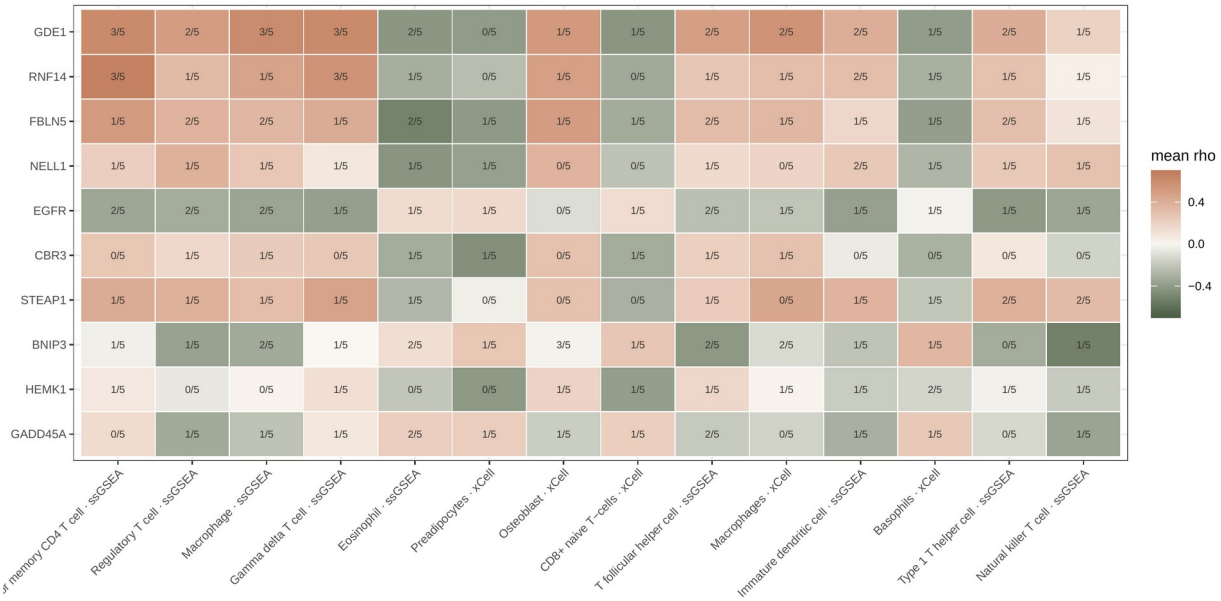


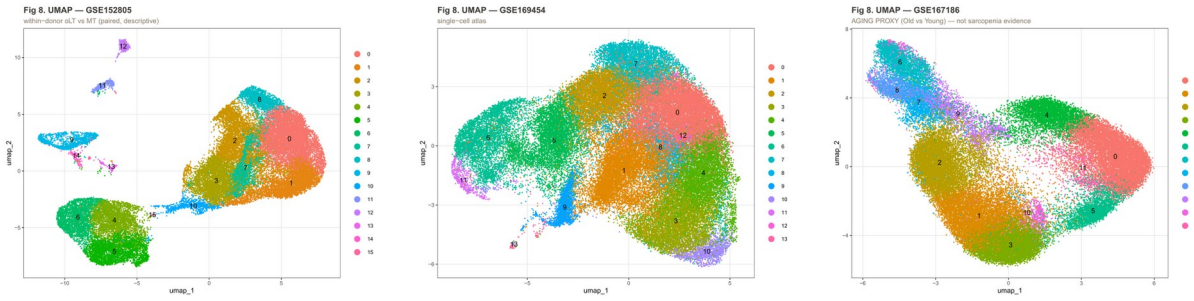
Fig 7c. Biomarker × cell−type Spearman correlations (OA training layer)

cell labels: n cohorts FDR<0.05 / n tested; columns ranked by mean |rho|

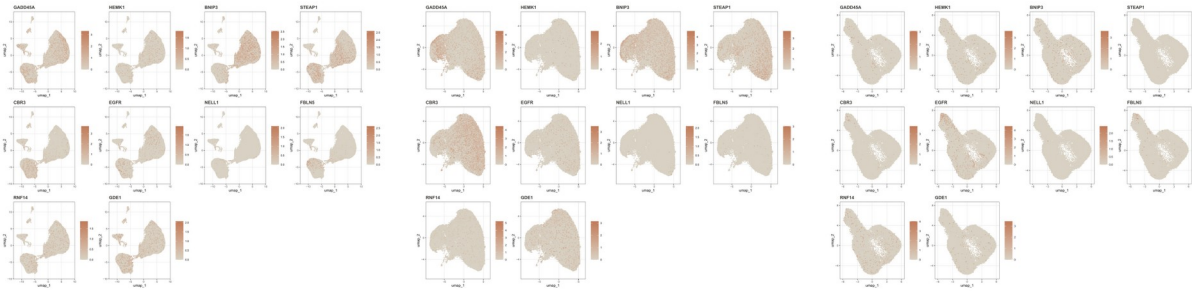


S-Fig 12 Single-cell and single-nucleus transcriptomic localization of the 10 biomarkers. Three datasets were processed with the Seurat v5 pipeline (quality control, normalization, Harmony integration, clustering, and dual-convention cell-type annotation): GSE152805 (OA cartilage scRNA-seq, 35,503 cells, annotated into 7 types), GSE169454 (chondrocyte scRNA-seq, 65,042 cells, 5 types), and GSE167186 (vastus lateralis single-nucleus RNA-seq, 52,269 nuclei, 6 types, aging proxy). (a) UMAP clustering views after Harmony integration for each dataset (points colored by cluster number; cluster–cell-type correspondence is given by the cell-type axis in (c) and the repository annotation table); (b) FeaturePlot expression distributions of the 10 biomarkers; (c) biomarker × cell-type localization dot plot (dot size = proportion of cells with detectable expression; color = mean expression); (d) donor-level pseudobulk biomarker comparison (log2FC/delta); (e) UCell module scores for the concordant-track and the two mirror-track gene sets with group comparison (score coverage 27/27, 37/37, 34–35/35; figure files archived at `results/moduleG/G2/<GSE>/`). Compartment-preference percentages denote the proportion of cells of the corresponding cell type with detectable expression of the gene (normalized RNA-layer expression greater than 0); differential testing used donor-level pseudobulk as the minimum standard. Dataset analysis-scope restrictions: GSE152805 was limited to the within-donor descriptive comparison of oLT and MT (n=3, with no case–control statement); the GSE167186 single-nucleus side is an aging proxy (Old versus Young comparison), with no significance after BH correction, reported as negative per the aging-proxy positioning.

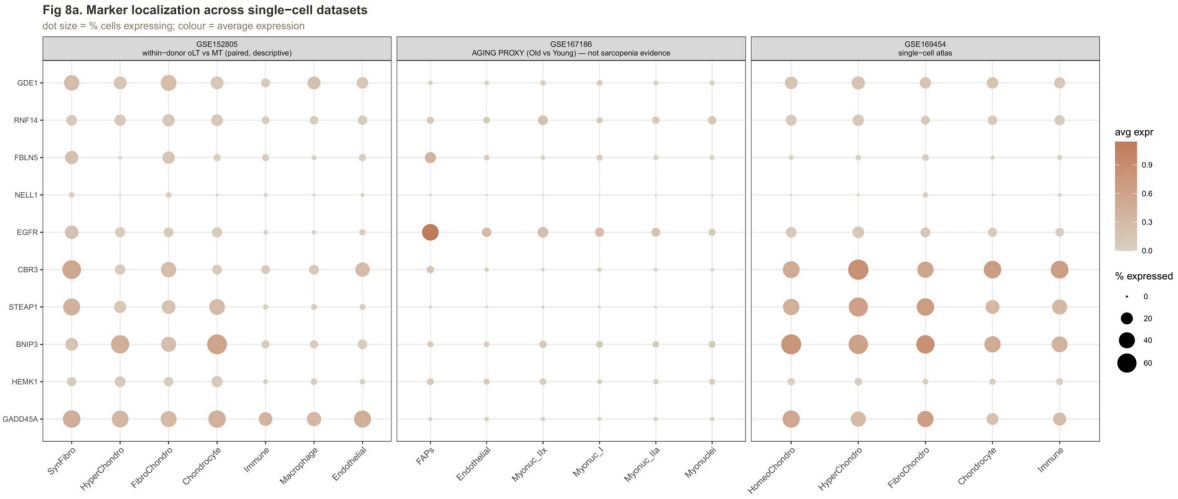
(a) Harmony-integrated UMAP clusters



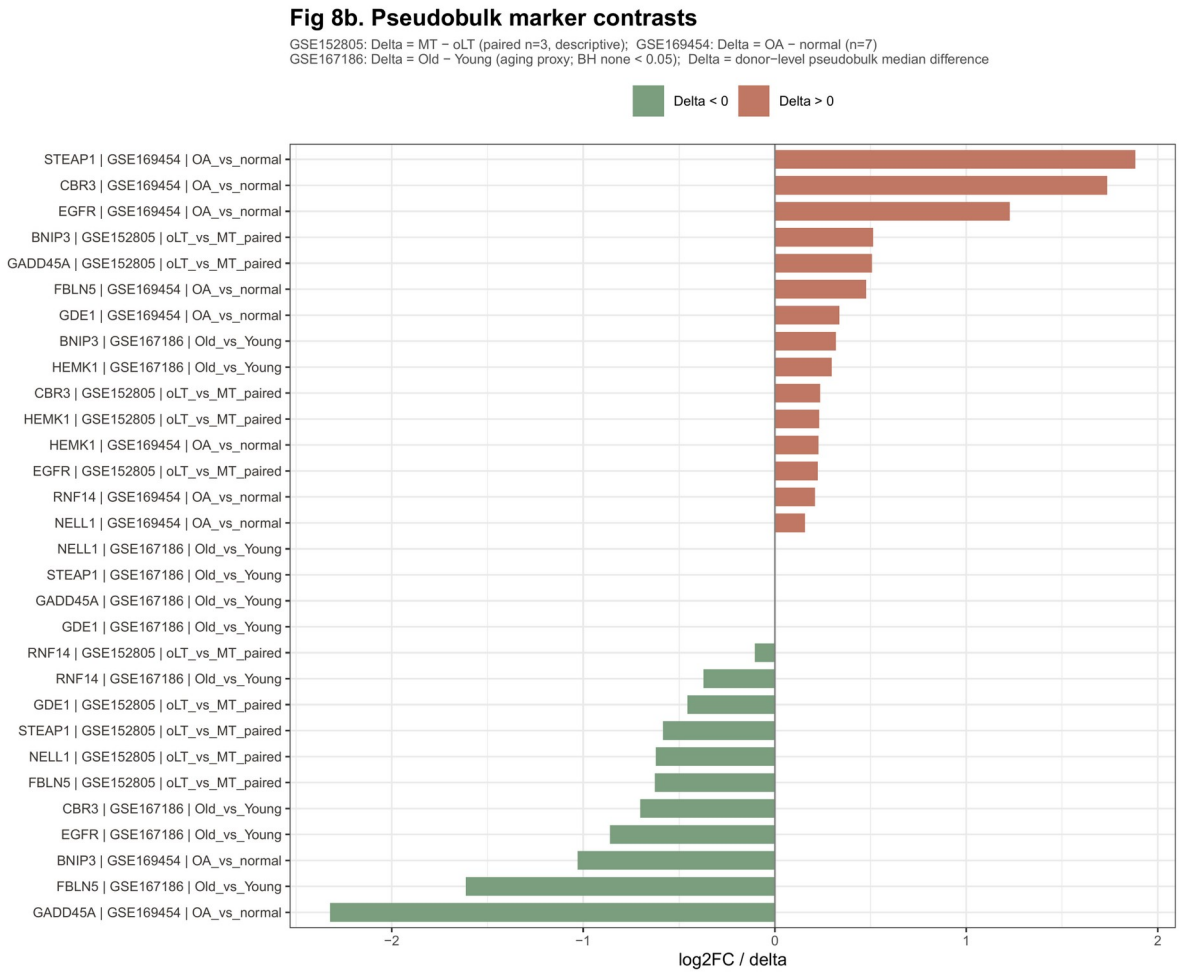
(b) FeaturePlot of the 10 biomarkers



(c) Biomarker × cell-type localization dotplot



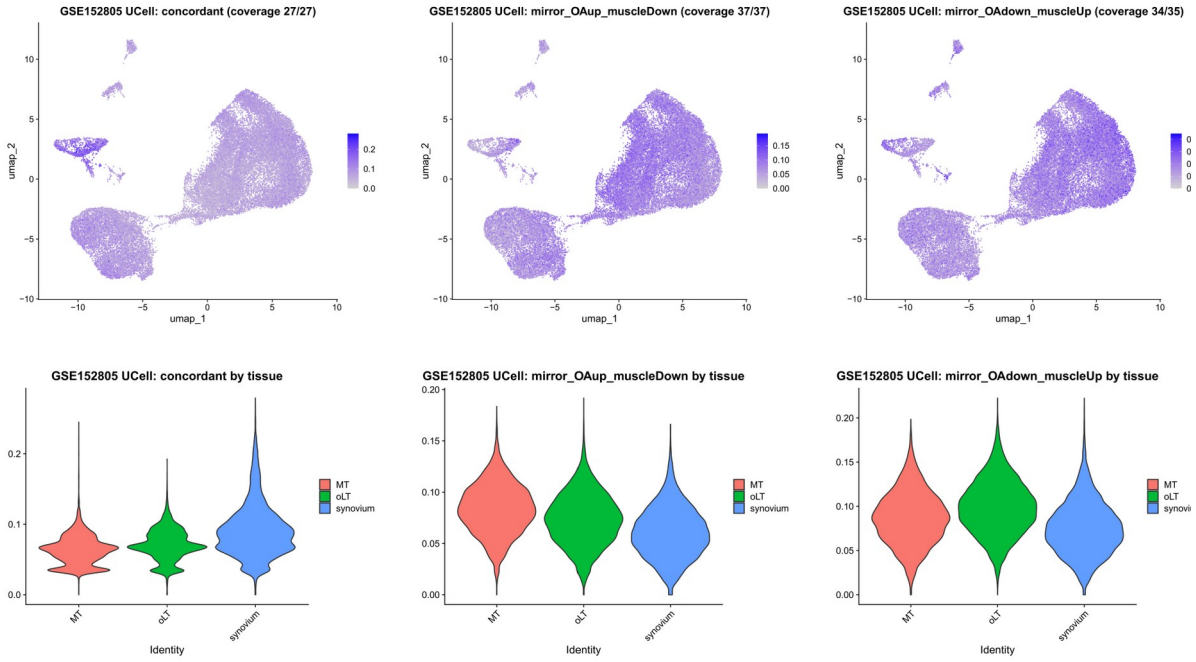
(d) Donor-level pseudobulk comparison



(e) UCell module scores of the three gene-set tracks

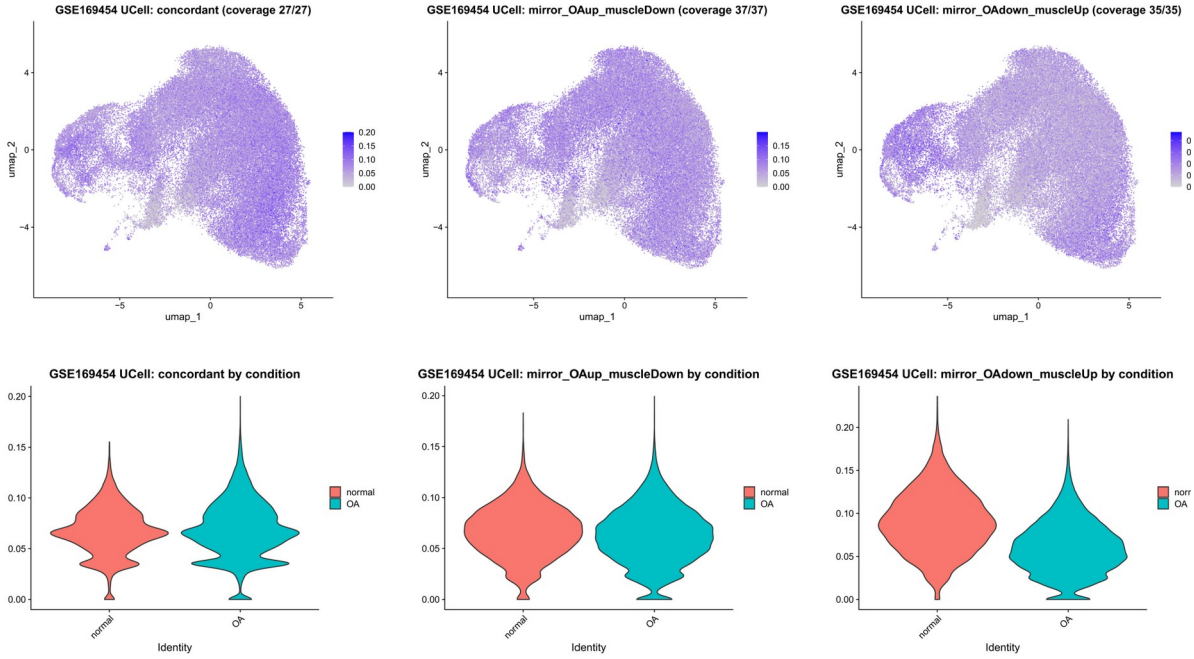
GSE152805 — OA cartilage scRNA-seq (VlnPlot by tissue)

concordant | mirror: OA-up / muscle-down | mirror: OA-down / muscle-up



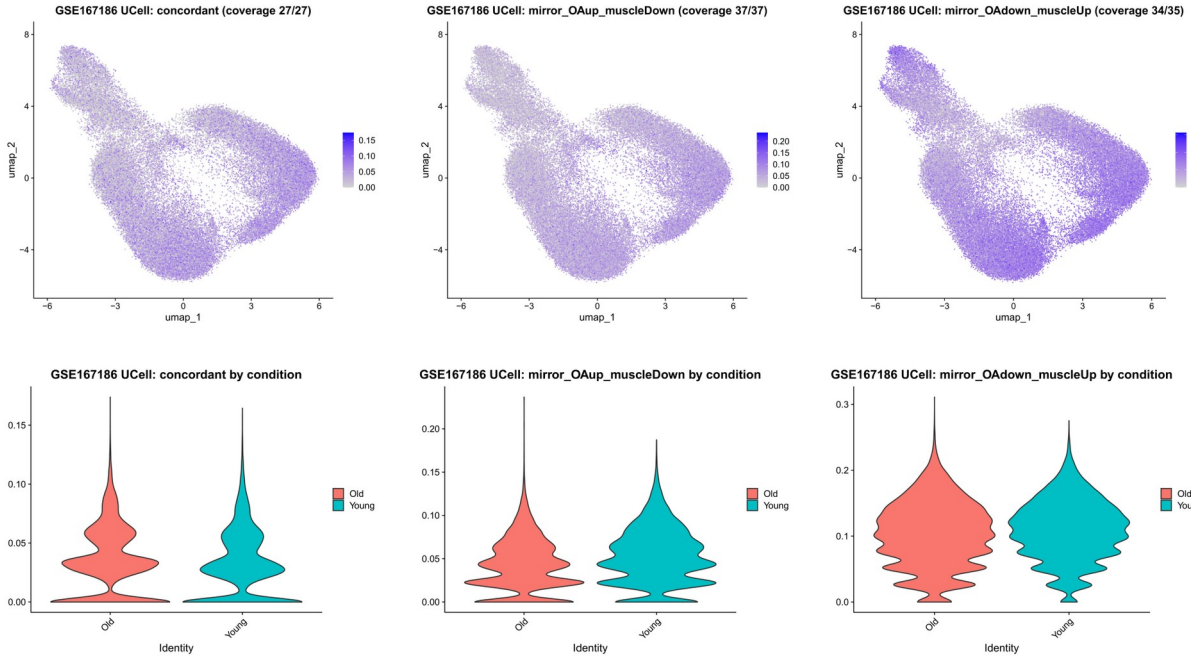
GSE169454 — chondrocyte scRNA-seq (VlnPlot by condition)

concordant | mirror: OA-up / muscle-down | mirror: OA-down / muscle-up



GSE167186 — muscle snRNA-seq, aging proxy (VlnPlot by condition)

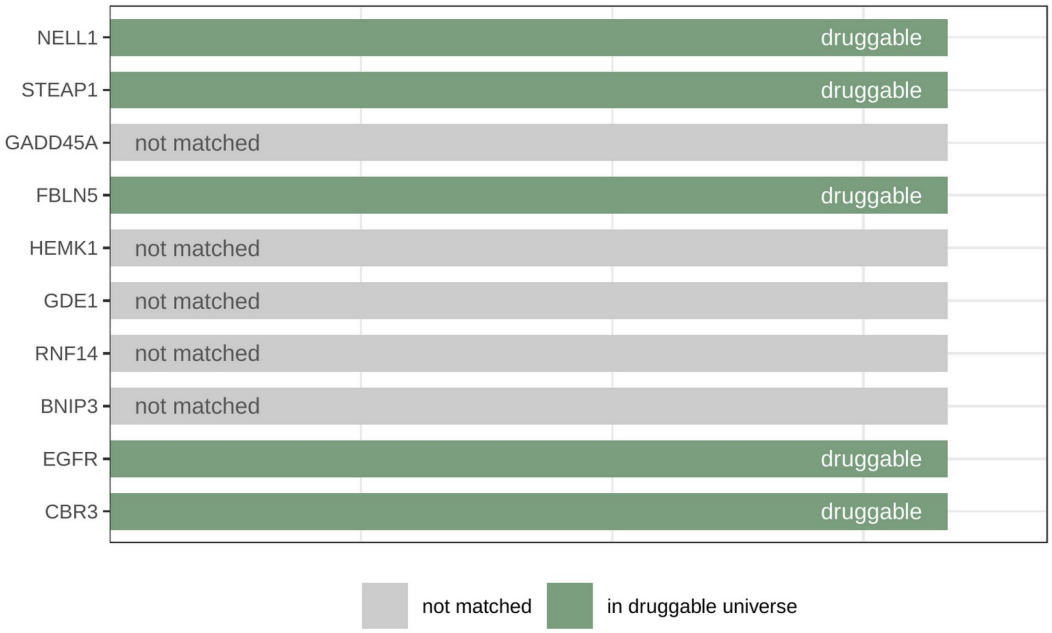
concordant | mirror: OA-up / muscle-down | mirror: OA-down / muscle-up



S-Fig 13 Translational extension: druggable-genome mapping and candidate-drug enrichment. (a) Mapping of the 10 biomarkers against the full druggable gene set (markers_only tier): the druggable list was the union of the Finan et al. list (4,463 genes) and the DGIdb 4.0 “druggable genome” category (5,809 genes) (5,928 genes in total); 5/10 biomarkers (NELL1, STEAP1, FBLN5, EGFR, CBR3) were hits; the Finan-only sensitivity analysis likewise gave 5/10. (b) Nominally significant candidates from DSigDB candidate-drug enrichment (561 of 4,026 entries entered testing; no entry passed FDR correction; per the pre-specified downgrade scheme, the top 5 nominally significant candidates are reported, exploratory): gedunin, dasatinib, celastrol, daunorubicin, and pyrrolidine dithiocarbamate (P between 6.4×10^{-4} and 3.0×10^{-3}). All candidate-drug evidence above derives from other indications; transfer to OA/sarcopenia is hypothetical in nature and serves only as drug-repurposing leads; the complete summary is provided in Table S18.

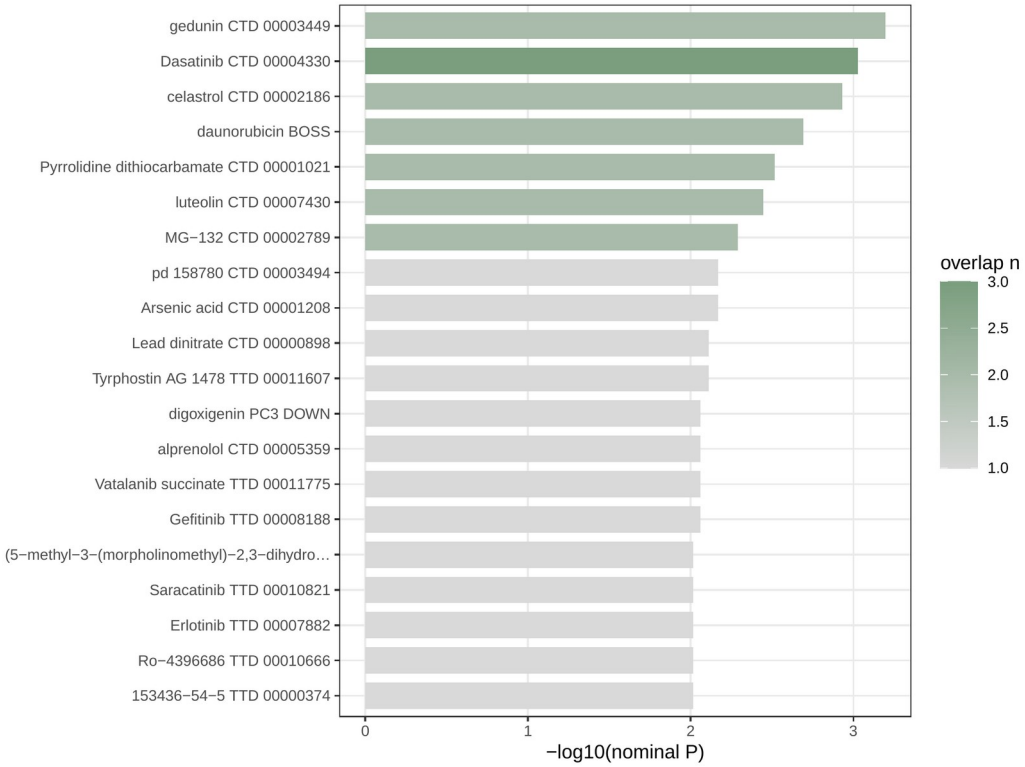
S−Fig 13a. Druggable−genome overlap (markers−only)

5/10 biomarkers in Finan & DGIdb druggable union (n = 5,928)



S−Fig 13b. DSigDB drug perturbation candidates

exploratory ORA; top 20 by nominal P; no term passed FDR (BH, 561 tested)



2. Supplementary table notes (Tables S1–S28)

Table S1 Summary of Module A dataset preprocessing and quality-control status. This table is the final run-status table of Module A, recording, dataset by dataset, the processing status of the 21 formally included datasets (18 bulk transcriptomic and 3 single-cell/single-nucleus datasets); the final status was 21/21 OK, excluding GSE129147, which had been removed from the main analysis. The Module A single-cell/single-nucleus datasets were registered only and handed over to Module G; the per-sample phenotype tables (`{GSE}_pheno.csv`) and QC plots (`{GSE}_QC.pdf`) of the bulk datasets are supporting files, and platforms, measured sample sizes, and role assignments follow Table S16. The key inclusion/exclusion conventions (exclusion of rheumatoid arthritis samples, removal of 4 UNCLASSIFIED samples in GSE167186, baseline-only sampling for follow-up samples, restriction of GSE12021 to GPL96, and use of GSE169077 pooled samples for direction validation only) are the same as in the main-text Methods. Source file(s): `moduleA_status_2026-08-22_final.csv` (21 data rows, 21/21 OK; SHA-256 identical to the uploaded copy `moduleA_status_2026-08-22(1).csv`). The older `moduleA_status_2026-08-22.csv` is a historical file from the initial run containing ERROR and does not serve as the submitted Table S1.

Table S2 Full results of per-cohort differential expression analysis. This table compiles the genome-wide case–control differential expression results for 10 cohorts, each containing complete statistics including logFC, t value, P value, and adj.P. Differential analysis in the OA training layer was performed with limma and DESeq2; per-cohort significant genes (adj.P < 0.05 and |log2FC|≥0.585, approximately 1.5-fold; descriptive convention) numbered 445–3,423 per cohort, and the RRA input consisted of whole-genome ranked two-sided lists. The muscle-side confirmed sarcopenia cohorts showed nearly 0 differentially expressed genes under FDR correction, a weak-signal landscape; therefore, per the prespecified rule, a nominal P < 0.05 cutoff combined with RRA direction-consistency filtering was applied. For GSE51588, same-donor two-site samples were handled with duplicateCorrelation to account for within-donor correlation, and the group contrasts for the aging-proxy cohorts were executed according to the prespecified CONTRAST_OVERRIDE. Source file(s): `results/moduleB/layer1_DE_GSE*.csv`, 10 cohort files in total

(GSE111006/GSE111010/GSE111016/GSE114007/GSE12021/GSE1428/GSE25941/GSE51588/GSE55235/GSE55457).

Table S3 Master table of RRA cross-cohort integration and dual-track candidate genes. This table lists all intermediate products of cross-cohort integration and dual-track construction: the OA-side per-cohort whole-genome ranked two-sided lists were integrated by RRA, yielding 511 integrated significant genes at adjusted adj.P < 0.05; intersection of the two sides yielded 99 shared candidate genes, classified by the directional relationship of dysregulation in the two diseases into the concordant track (27 genes) and the mirror track (72 genes, comprising 37 OA-up/muscle-down and 35 OA-down/muscle-up), with a mirror fraction of 72.7% (one-sided binomial test $P=3.5\times10^{-6}$). A sensitivity analysis reconstructed using only the confirmed sarcopenia cohorts is also included: 21 candidates were obtained, with a mirror fraction of 76.2% (16/21), of which 15 overlapped with the main-analysis mirror track (93.8% (15/16)); leave-one-dataset-out (LODO) and donor-overlap merge correction sensitivity analyses of the integrated set are presented in Tables S19 and S20, and their fully corrected combination (merged OA lists × anchor-only cohorts) in Table S21 — membership-level stability was moderate and concentrated at threshold edges, whereas direction-level mirror dominance held across all testable sensitivity analyses (65.1%–88.1%, all P<0.05); phenotype-tier stratification, module-aware permutation testing, and muscle-side threshold sensitivity of the mirror fraction are reported in Tables S22–S24. The genome-wide signed rank correlation ρ=−0.058 serves only as global corroboration; it derives from the same data source used to define the mirror track and does not constitute independent evidence. Source file(s): `results/moduleB/layer2_RRA_OA.csv` (511 robust genes), `layer2_RRA_muscle.csv`, `layer2_RRA_muscle_sarcopenia_only.csv`, `layer2_shared_DE_genes.csv` (99 candidates), `layer2_sensitivity_sarcopenia_only.csv` (76.2% convention), `CORE_concordant_genes.csv`, `CORE_mirror_genes.csv`; LODO products `results/moduleB/LODO/`, donor-overlap merge correction products `results/moduleB/LODO_merged/`.

Table S4 WGCNA module membership and high-confidence gene table. This table lists the module membership results of weighted gene co-expression network analysis (soft threshold selected by $R^2 > 0.8$): 8 disease-related modules were identified on the OA side (2,119 high-confidence genes), and 4 modules were identified on the muscle side based on GSE111016 (soft

power=8; 304 high-confidence genes); after mapping dual-track genes to disease-related modules, 7/27 concordant-track and 27/72 mirror-track genes fell within high-confidence modules. This layer is positioned as a confidence annotation rather than a hard-intersection filter and does not alter dual-track membership. Source file(s): `results/moduleB/layer3_WGCNA_modules_OA.csv` (8 modules / 2,119 high-confidence genes) and `results/moduleB/layer3_WGCNA_modules_muscle.csv` (GSE111016, soft power=8, 4 modules / 304 genes).

Table S5 Machine-learning feature selection and biomarker details. The feature space was the three-way intersection of the 99 dual-track genes, the genes detectable in each training cohort, and the genes detectable on the GPL6947 platform (the GSE48556 blood platform) (84 genes; platform information was available a priori); the training set comprised 5 OA cohorts merged after per-cohort z-score standardization, totaling 117 samples (65 cases/52 controls), with GSE51588 deduplicated by donor (reduced from 50 to 20 samples). This table lists the complete records of independent selection by four algorithms and their cross-voting (vote-count convention 4/4/4/3×6/2); under the strategy of prioritizing the LASSO ∩ SVM-RFE intersection, supplemented by random forest and XGBoost top-10 votes, the 10 biomarkers were locked (NELL1, STEAP1, GADD45A, FBLN5, HEMK1, GDE1, RNF14, BNIP3, EGFR, CBR3), including each biomarker's track membership (5 concordant track, 3 OA-down/muscle-up mirror track, 2 OA-up/muscle-down mirror track) and logistic coefficient direction (10/10 consistent with the discovery-layer dysregulation direction); the full values of exact linear SHAP global importance (logit scale, training-mean baseline) are also provided, with mean|SHAP| ranked from GADD45A (2.43) at the top to GDE1 (0.24). Source file(s): `results/moduleD_v2/feature_selection_votes.csv`, `feature_votes_by_algorithm.csv`, `moduleD_v2_biomarker_table.csv`, `moduleD_v2_SHAP_importance.csv`, `moduleD_v2_SHAP_values.csv`.

Table S6 Full table of Mendelian randomization sensitivity analyses. This table lists the complete sensitivity analysis results for the 32 actually executed cis-instrument MR tests, including the Steiger directionality test (32/32 all passed), Cochran's Q heterogeneity statistics, MR-Egger intercepts, and MR-PRESSO pleiotropy test results, serving as the basis for the main-text judgment of "no pleiotropy evidence altering the conclusions." The multiple-testing

convention was Bonferroni as primary with BH as sensitivity, and the test family comprised 32 tests. Source file(s): `results/moduleH/moduleH_MR_all_methods.csv`.

Table S7 Full statistics table of external validation. This table compiles all external validation statistics according to the prespecified three-tier architecture of "cartilage primary validation — blood exploratory — muscle-proxy exploratory." **Cartilage primary validation layer**: GSE57218 (7 healthy vs 33 OA) AUC=0.818 (95%CI 0.614–1.000, DeLong method); the AUC > 0.7 performance claim applies only to this layer; GSE168505 (n=7) AUC=0.750, reported with down-weighting due to the extremely small sample size; GSE206848 was not validated per the established rule because biomarker platform coverage was 0/10 (not a biological negative). The external calibration slope was 0.268 and the intercept 5.45 (the convention of benchmarking external calibration against a constant model is explained in the main text); decision curve analysis (DCA) showed no net-benefit interval simultaneously superior to both the "treat-all" and "treat-none" strategies, consistent with the high prevalence of this cohort (82.5%), and the external evidence is primarily discrimination-based. **Blood exploratory layer**: GSE48556 (PBMC, 106 OA/33 controls) AUC=0.605 (0.499–0.713), with direction concordance 6/10; this layer is positioned as exploratory with no performance threshold; the blood sub-signature nested cross-validation (with feature re-selection within folds) yielded OOF AUC=0.82 (0.711–0.92), with within-fold stable top-5 genes CALML4/DCLRE1C/GADD45A/KLF2/NFIC, labeled hypothesis-generating and not constituting a diagnostic claim. **Muscle-proxy exploratory tier**: across the three frailty/aging proxy cohorts, the overall direction concordance rate was 9/20 (45%), below the 50% chance expectation (two-sided binomial P = 0.82), and under the prespecified ≥ 2-cohort acceptance criterion only 2/10 (GDE1, BNIP3) were accepted; this tier therefore provided no supportive evidence and is reported as directions and leads only; GSE136344 had an empty intersection with the biomarkers and contributed no direction statistics. The only gene-level discrepancy between direction acceptance 7/10 (cohort-level z-score between-group mean convention) and the single-gene AUC layer 6/10 (rank-based convention) is GDE1, which this study counts as directionally discordant at the single-gene layer under the conservative (rank/AUC) convention. Source file(s): `results/moduleE/` (`moduleE_cartilage_AUC.csv`, `moduleE_cartilage_direction_by_gene.csv`, `moduleE_muscle_mirror_direction.csv`, `moduleE_biomarker_verdict.csv`) + `results/moduleD_v2/`

(calibration_internal_cv_multicohort.csv, calibration_external_blood_GSE48556.csv, dca_internal_cv_multicohort.csv, dca_external_blood_GSE48556.csv, direction_check_GSE48556.csv) + blood exploratory layer blood_exploratory/EXPLORATORY_blood_fold_selection_stability.csv and EXPLORATORY_blood_DE_dualtrack_genes.csv.

Table S8 Two-sided known-causal gene control list (Appendix D list). This table compiles the OA-side 13-gene known-causal list (MAPK3, CSK, IFNGR2, MVD, FES, ITGA2, GZMK, LGALS3, ITIH1, DNAJB12, USP8, IL12B, RGMB; inclusion criteria were tier 1/2 in the respective papers or author-designated core targets, without exhaustive enumeration; LGALS3 is tier 2 and was retained per the 2026-08-29 adjudication) and the sarcopenia-side 6-gene list (HP, HLA-DRA, MAP3K3, MFGE8, COL15A1, AURKA; Yin et al. 2024 JCSM), and verifies one by one that none overlaps with the 10 biomarkers of this study (GADD45A, EGFR, NELL1, STEAP1, FBLN5, RNF14, HEMK1, GDE1, CBR3, BNIP3), supporting that the candidates of this study are novel signals under the two-disease shared framework rather than replications of existing single-side findings. Source file(s): results/moduleH/moduleH_known_causal_overlap.csv.

Table S9 Master table of the CIBERSORT dual-version comparison. This table presents the per-combination comparison between the open-source dicepro implementation and an independent rerun on the official Stanford CIBERSORTx platform, comprising 88 rows (4 high-confidence cohorts GSE55235/GSE55457/GSE12021/GSE48556 × 22 LM22 cell types), including case/control means, deltas, P values, adjusted P values, and direction-consistency determinations for both versions. The overall direction concordance rate was 92.0% (81/88); the Spearman ρ of the two-version delta effect sizes across the four cohorts was 0.862–0.992 (all $P<10^{-6}$); the 3 items significant in both versions were fully concordant in direction, and the complete details of the 3 single-side-significant items and the 7 zero-value noise direction conflicts are faithfully listed; a scatter plot comparing the two-version deltas is attached. Source file(s): fix15_双版对照_CIBERSORTx官方_vs_dicepro_20260903.csv (88 data rows, verified); supporting report fix15_双版对照报告_20260903.md.

Table S10 List of analysis tools and R package versions. This table lists, module by module, all R packages used in the analyses and their version numbers (60 rows, 54 packages,

including magick/pdftools/png used in the figure redrawing and composition stage of scripts 22–38); all analyses were completed in the R 4.6.0 environment; the table was finalized (frozen) by an actual run of `15b_sessionInfo_S10工具版本表_封版_20261008.R` on 2026-10-08, with all 54 packages installed and versions as measured on the local machine, and the full sessionInfo archived in the same directory. This table was finalized (frozen) in synchronization with the GitHub repository (https://github.com/Chiaki74/OA-sarcopenia-dual-track; full analysis pipeline and frozen supplementary tables first archived on 2026-10-12, archive commit 34fbbdc; results archive updated 2026-10-19, commit 501ceac); if the R environment is upgraded before submission, rerunning 15b will update it. Source file(s): generated together with the `sessionInfo` output accompanying the code repository (generating scripts `15_sessionInfo_S10工具版本表_20260915.R` and its frozen rerun `15b_sessionInfo_S10工具版本表_封版_20261008.R`, both archived under `scripts/` in the code repository).

Table S11 Single-gene diagnostic performance table of the 10 biomarkers. This table lists the single-gene AUCs of the 10 biomarkers under the dual conventions of the training layer and the GSE57218 external layer: the training layer ranged from 0.636 (HEMK1) to 0.851 (GDE1); in the external layer, AUCs and 95%CIs (DeLong method) are reported after explicitly fixing the direction to the discovery-layer direction. In the external layer, 6/10 were directionally concordant (GADD45A and EGFR highest, both with AUC 0.944) and 4/10 were reversed (HEMK1 most strongly reversed, direction-fixed AUC=0.152); per the prespecified convention, the single-gene external validation results do not support any performance claim. Source file(s): 表`S11_单基因诊断效能_方向固定订正版_20260916.csv` (10 data rows, verified; the historical draft `moduleI_single_gene_AUC.csv` has been superseded by the direction-fixed corrected version).

Table S12 Robustness summary of the seven-classifier panel. This table lists, for the seven classifiers (the glm baseline plus six complementary learners) under the same training set and cross-validation scheme, the repeated 3 × 10-fold cross-validation AUCs (0.957 kNN – 0.981 radial-kernel SVM) and the GSE57218 external AUCs (0.788 nnet – 0.887 naive Bayes), together with the explicitly fixed random seeds for each method (20260914–20260921) to facilitate reproduction. The glm row corresponds to the main-model logistic regression baseline (external AUC 0.818, consistent with the main text) and is retained as an algorithmic comparator reference; the originally planned xgbTree could not be executed because of interface

incompatibility between caret 7.0.1 and xgboost 3.2.1.1 (confirmed by the 2026-09-13 script 12e diagnostic, including an nthread=1 fix attempt; declared in the methodological main text), and the finalized (frozen) table no longer retains its empty row. Source file(s): 表 S12_七分类器稳健性汇总_封版_20261008.csv (7 data rows; finalized (frozen) by deleting the xgbTree empty row from the Module I product moduleI_classifier_robustness.csv, with no changes to any values, verified item by item against the 2026-09-13 run log).

Table S13 Full table of nomogram bootstrap validation. This table lists all statistics of the bootstrap internal validation (B=200) of the 10-biomarker logistic nomogram: apparent C-index=0.989, optimism-corrected C-index=0.97 (corrected Dxy=0.940), and corrected calibration slope 0.394, indicating marked overfitting in the internal evaluation; a coefficient-direction consistency check between the lrm and glm implementations for the 10 biomarkers is also included (10/10 concordant). **Supplementary table S13-Appendix** provides the complete main-model equation: 10 coefficients plus the intercept, ORs and Wald 95%CIs, Wald z and P values, and direction annotations (表 S13 附_主模型完整系数表_20261003.csv; deterministically refitted from the 2026-09-16 version of the 117-sample training matrix via 26_moduleI_主模型系数提取_20261003.R, with apparent C-index=0.989 consistent with the main text). The terms with P<0.05 are STEAP1, GADD45A, HEMK1, and CBR3. Source file(s): moduleI_nomogram_validate.txt (Module I product, rms::validate output, B=200, values verified as consistent).

Table S14 TF–miRNA upstream regulatory edge table and hub summary. This table lists 75 predicted upstream regulatory edges for the 10 biomarkers (73 transcription-factor edges and 2 miRNA edges), covering 6/10 biomarkers; NELL1, STEAP1, RNF14, and CBR3 have no recorded significant upstream regulators; each edge is annotated with an evidence tier (nominal_only or multi-source high-confidence), and a hub summary is appended (no regulator targets ≥3/10 biomarkers). Source file(s): results/moduleJ/moduleJ_TFmiRNA_edges.csv (75 edges) and results/moduleJ/moduleJ_TFmiRNA_hub.csv (hub summary); generating scripts 13_moduleJ_TFmiRNA 调控网络_20260914.R and 13b_moduleJ_补丁_TF 库重试与 TRRUST 兜底_20260914.R, archived under scripts/ in the code repository — the cited results/ tables are deposited in the repository (results archive, commit 501ceac); only large cache files (*.rds) are excluded per the repository .gitignore policy and are regenerated by the pipeline.

Table S15 Master table of CellChat biomarker-related ligand–receptor interactions and summary of per-dataset between-group comparisons. This table compiles the CellChat cell–cell communication analysis results for the three datasets GSE169454, GSE152805, and GSE167186: the number of L–R interactions in OA cartilage increased relative to normal controls (GSE169454: 505 vs 69; GSE152805 within-donor: 1,761 in the MT region and 282 in the oLT region); the main analysis layer detected no significant interactions involving the 10 biomarkers as ligands/receptors; GSE167186 follows the aging-proxy convention, and its EGF→EGFR interaction serves only as an exploratory echo and does not support a sarcopenia conclusion. Source file(s): `results/moduleK/moduleK_biomarker_LR_pairs.csv` (biomarker-related L–R table) and the per-dataset between-group summaries `results/moduleK/moduleK_*_compare_summary.csv`; generating script `14c_moduleK_CellChat_分组标签与presto修正_20260914.R` (finalized correction of the 14/14b chain), archived under `scripts/` in the code repository — the cited `results/` tables are deposited in the repository (results archive, commit 501ceac); only large cache files (*.rds) are excluded per the repository .gitignore policy and are regenerated by the pipeline.

Table S16 Master table of the 21 datasets. This table lists the GEO accession, tissue source, platform, measured sample size, analysis role, and three-way phenotype classification of all 21 public datasets integrated in this study (18 bulk transcriptomic and 3 single-cell/single-nucleus datasets); among them, GSE48556 was measured as 129 female/10 male (the “all-female” description is not used), and GSE167186 comprises 72 bulk plus 17 single-nucleus samples, totaling 89; it is counted once among the 21 datasets for its single-nucleus component, and its 72 bulk records were not analysed in any layer (bulk differential-expression inputs comprise the ten cohorts listed under Table S1, which do not include GSE167186). The access date for all GEO datasets was August 2026. Source file(s): 表`S16_21`数据集总表_最终工作版`_20261003.csv` (21 data rows; derived from 数据集信息核对表_预填版`_20260813.csv` by removing GSE129147, which had been removed from the main analysis, and implementing, per main text v13 and the decision memorandum, the conventions of GSE48556 129 female/10 male and the final GSE167186 n=89; a review version 表`S16_21`数据集总表_最终工作版`_20261003.xlsx` is provided in parallel).

Table S17 Summary of MR and colocalization. This table compiles the main MR results and colocalization analyses: of 80 candidate combinations (10 biomarkers × 4 outcomes × 2 eQTL sources), 32 were actually executed because cis instruments could be constructed, constituting the multiple-testing family (Bonferroni as primary and BH as sensitivity; none significant after correction); 6 nominally significant candidates are listed (FBLN5–appendicular lean mass IVW P=0.0033, CBR3–knee OA IVW P=0.0084, HEMK1–appendicular lean mass GTEx-source Wald P=0.010, GDE1–low grip strength IVW P=0.029, HEMK1–appendicular lean mass eQTLGen-source IVW P=0.033, RNF14–walking pace IVW P=0.046). The colocalization section lists abf posterior probabilities for 24 combinations (primary threshold PPH4>0.8, p12=5×10^{-6}) together with SuSiE fine-mapping sensitivity results. Source file(s): `results/moduleH/moduleH_MR_summary.csv` + `results/moduleH/moduleH_coloc_summary.csv` (for SuSiE sensitivity see also `results/moduleH/fix11_susie/fix11e_susie_summary_convergence.csv`).

Table S18 Summary of druggable targets/candidate drugs. This table compiles the translational extension analysis: 5 of the 10 biomarkers hit the druggable genome union (5,928 genes; merging the Finan et al. list and DGIdb 4.0) — NELL1, STEAP1, FBLN5, EGFR, CBR3 — with the Finan-only sensitivity analysis likewise yielding 5/10 and an evidence tier of markers_only; of 4,026 DSigDB candidate-drug enrichment entries, 561 entered testing and none passed FDR correction; per the prespecified downgrade plan, the nominally significant top-5 candidates are reported (exploratory): gedunin (P=6.4×10^{-4}), dasatinib (9.4×10^{-4}), celastrol (1.2×10^{-3}), daunorubicin (2.0×10^{-3}), pyrrolidine dithiocarbamate (3.0×10^{-3}). Source file(s): `results/moduleI/moduleI_druggable_overlap.csv` + `results/moduleI/moduleI_dsigdb_enrichment.csv`.

Table S19 Full-cohort leave-one-dataset-out (LODO) robustness matrix. Leaving out, in turn, 1 of the 5 OA training cohorts and 1 of the 5 muscle-side cohorts, the side-specific RRA and dual-track candidate sets were reconstructed (10 items in total), reporting retention of the side-specific significant set (retained count/retention rate/Jaccard), the genome-wide −log10(RRA score) rank correlation (Spearman ρ), the dual-track candidate count and retention rate, and the mirror-track retention rate and mirror fraction (one-sided binomial test). Key points: ① dual-track candidate retention was 1.0%–59.6%, with set changes concentrated at nominal-

threshold edges (the median leave-one-out nominal score of dropped candidates was approximately 0.10); ② ρ was 0.82–0.91 across all leave-one-outs, indicating a stable direction-level structure; ③ the mirror fraction was 65.1%–88.1% across all 8 testable leave-one-outs (all $P<0.05$); ④ removing GSE55457 or GSE12021 (the donor-overlap pair) collapsed the candidate set to 1 (the sole survivors were both mirror track), directly exposing the load-bearing role of the duplicated evidence; ⑤ GSE111006 (36 controls/4 cases) was the least disruptive muscle-side cohort (candidate retention 59.6%, the highest among the ten leave-one-outs) — it has 0 significant genes at the main-analysis threshold, yet 26 of the 29 (89.7%) adjusted-significant genes in the main-analysis muscle track were directionally concordant in this cohort (two-sided binomial test $P=1.5\times10^{-5}$). Source: 表 S19_LODO 全队列矩阵_20261003.csv; generating scripts 23_moduleB_LODO_GSE111006 敏感性_20261003.R, 24_moduleB_LODO 全队列矩阵_20261003.R.

Table S20 Donor-overlap correction sensitivity analysis (Stouffer merging of GSE55457+GSE12021). GSE55457 and GSE12021 exhibit donor overlap (noted in the main-text Methods), and their use as two independent lists in the main-analysis RRA constitutes duplicated evidence, violating the RRA list-independence assumption. In this table, the t statistics of the two cohorts were merged by the Stouffer method into a single study-level list ($Z=\Sigma t/\sqrt{k}$, with direction taken as sign(Z); the OA side changed from 5 lists to 4 independent lists), and the RRA and dual-track candidate sets were reconstructed: the OA-side integrated significant set shrank from 511 to 10 genes (HMGB2, JUN, JUNB, SIAH2, ZFP36, SAP30, GADD45A, MXI1, CCNH, KLF4; all downregulated, conserved across synovium and cartilage), and the dual-track candidates shrank from 99 to 1 — GADD45A (mirror track: OA-down, muscle-up; simultaneously a member of the biomarker panel and the top-ranked SHAP importance feature). In study-level leave-one-out under the merged structure, removing any single list left the OA-side significant set at 0, suggesting that the RRA correction power of 3–4 independent lists is near the floor and that the 10-gene set requires all lists to be present simultaneously. Conclusion positioning: the individual

membership of the integrated set is sensitive to the independence assumption, and the main-analysis results should be read as a candidate pool; the direction-level mirror-dominance conclusion is unaffected (see Table S19 and the main text). Source: 表 S20_供体重叠合并校正汇总_20261003.csv **(main table),** 表 S20 附_供体重叠合并校正_OA 侧 RRA 全表_20261003.csv, 表 S20 附_研究级留一_20261003.csv; **generating script** 25_moduleB_合并供体重叠列表敏感性_20261003.R.

Table S21 Fully corrected rebuild analysis (Stouffer-merged four-list OA side × anchor-only sarcopenia cohorts). Donor-overlap correction (Table S20) and anchor-only reconstruction (Table S3) had each been applied only in isolation; this table reports their combination as a parallel correction-tier analysis, as the most conservative reconstruction of the dual-track framework. The 2 × 2 grid (candidate count / mirror count / mirror fraction / one-sided binomial P) is: main analysis (5 OA lists × 5 muscle cohorts) 99 / 72 / 72.7% / P = 3.5×10^{-6}; donor-overlap correction only (merged 4 lists × 5 cohorts) 1 / 1 / 100% / P = 0.5; anchor-only (5 lists × 3 anchor cohorts) 21 / 16 / 76.2% / P = 0.013; full correction (merged 4 lists × 3 anchor cohorts) 1 / 1 / 100% / P = 0.5. The sole fully corrected survivor was GADD45A (mirror track, OA-down/muscle-up; OA RRA adj.P = 0.011; muscle nominal RRA score = 0.014), a member of the 10-marker panel and its top SHAP feature. Within the fully corrected structure, leave-one-out of any of the four OA-side lists (GSE114007, GSE55235, GSE51588, or the merged GSE55457+GSE12021) or of GSE111006 on the muscle side collapsed the candidate set to zero, whereas leaving out GSE111016 or GSE111010 retained GADD45A—the survivor is therefore directionally robust but threshold-fragile. Because the mirror fraction exceeded 50% in all four grid cells (72.7–100%) while gene-level membership does not survive simultaneous correction, mirror dysregulation is advanced strictly as a consistent, hypothesis-generating direction-level signature. Source file(s): `results/moduleB/fully_corrected/`完全校正_2x2 网格汇总.csv, 完全校正_双轨候选明细.csv, 完全校正_LODO 矩阵.csv, 完全校正_摘要.txt; generating script

45_moduleB_完全校正重建_合并OA四列表x锚定肌侧_20261014.R (run 2026-10-14; R 4.6.1, RobustRankAggreg).

Table S22 Block-level sign-flip permutation test of the mirror fraction. Because co-expression correlation violates the gene-level independence assumed by the binomial test, candidate genes were grouped into blocks defined by WGCNA co-expression modules (edges connect same-module pairs, with the unassigned grey module excluded; connected components form blocks), and the sign of each block was flipped jointly under the null (10,000 permutations). Main analysis (99 candidates): 45 blocks; observed mirror fraction 72.7%; strict block-flip empirical P = 0.2331; block majority-vote 34/45 (conservative binomial P = 4.1×10^{-4}). Anchor-only reconstruction (21 candidates): 11 blocks; mirror fraction 76.2%; empirical P = 0.31; majority-vote 9/11 (P = 0.0327). Interpretation: the gene-level binomial P (3.5×10^{-6}) is anti-conservative under module correlation, while the strict block-flip test is underpowered at this block count; the block majority-vote statistic remained significant in both structures. We therefore rest the direction-level claim on consistency across structures and thresholds (Tables S19–S21, S23) rather than on any single P value. Source file(s): `results/moduleB/robustness_mirror/`区组置换检验_汇总.csv, 区组置换_零分布.csv, 区组归属表.csv; generating script **46_moduleB_镜像比例区组置换检验_与肌侧阈值敏感性_20261014.R** (run 2026-10-15; R 4.6.1).

Table S23 Muscle-side threshold sensitivity grid. The main analysis used a nominal muscle-side threshold of P < 0.05 (a data-driven choice; see main-text Methods); this grid reconstructs the dual-track candidate set across alternative thresholds (candidate count / mirror count / mirror fraction / one-sided binomial P). Five-cohort muscle side: P < 0.01: 43 / 33 / 76.7% / P = 3.0×10^{-4}; P < 0.02: 68 / 51 / 75.0% / P = 2.2×10^{-5}; P < 0.05 (main analysis): 99 / 72 / 72.7% / P = 3.5×10^{-6}; P < 0.10: 157 / 109 / 69.4% / P = 6.3×10^{-7}; P < 0.20: 220 / 147 / 66.8% / P = 3.4×10^{-7}. Anchor-only (three confirmed-sarcopenia cohorts): P < 0.01: 3 / 3 / 100% / P = 0.125; P < 0.02: 10 / 9 / 90.0% / P = 0.0107; P < 0.05: 21 / 16 / 76.2% / P = 0.0133; P < 0.10: 47 / 36 / 76.6% / P = 1.7×10^{-4}; P < 0.20: 77 / 52 / 67.5% / P = 1.4×10^{-3}. The mirror fraction exceeded 50% in all 10 cells (range 66.8%–100%) and was nominally significant in 9 of 10, indicating that the direction-level conclusion is not an artifact of the chosen muscle-side threshold. Source file(s): `results/moduleB/robustness_mirror/`肌侧阈值敏感性.csv;

generating script **46_moduleB_镜像比例区组置换检验_与肌侧阈值敏感性_20261014.R** (run 2026-10-15; R 4.6.1).

Table S24 Muscle-side phenotype-tier stratification and anchor-support decomposition of the mirror fraction. To quantify whether mirror dominance depends on phenotype definition, the muscle side was reconstructed in three tiers (candidate count / mirror count / mirror fraction / one-sided binomial P): T1 anchor-only (GSE111016/GSE111010/GSE111006, confirmed sarcopenia by original-study definitions): 21 / 16 / 76.2% / P = 0.0133; T2 anchor + aging-proxy (adding GSE1428/GSE25941, old versus young muscle; identical to the main analysis): 99 / 72 / 72.7% / P = 3.5×10^{-6}; T3 anchor + aging-proxy + frailty-proxy (adding GSE117525/GSE144304, frailty versus non-frail controls; genome-wide limma DE computed de novo under the same pipeline—GSE117525: 134 samples, 4,722 nominal $P < 0.05$ genes; GSE144304: 54 samples, 1,337): 163 / 107 / 65.6% / P = 4.0×10^{-5}. The mirror fraction thus exceeded 50% in every phenotype tier, with modest attenuation as broader proxy phenotypes were added; mirror-track overlap across tiers was 15/16 (T1 ∩ T2) and 54/72 (T2 ∩ T3). GSE136344 was not included in T3 because its usable contrast is metabolic syndrome versus elderly controls, which does not constitute a frailty/sarcopenia comparison. Anchor-support decomposition of the 99 main-analysis candidates (per-gene count of cohorts with nominal $P < 0.05$ and track-consistent direction): among the 42 candidates with support in ≥ 1 confirmed-sarcopenia cohort the mirror fraction was 81.0% (34/42; P = 3.4×10^{-5}), versus 71.4% (20/28; P = 0.018) among genes supported only by aging-proxy cohorts and 62.1% (18/29; P = 0.132) among genes without per-cohort nominal support (RRA-integrated only). Mirror-track genes drew anchor-cohort support more often than concordant-track genes (47.2% vs 29.6%). Mirror dominance is therefore strongest precisely where confirmed-sarcopenia evidence exists and is not an artifact of the aging-proxy cohorts. Replication check: the T1/T2 reconstructions exactly reproduced the frozen main-analysis values (21 and 99 candidates). Source file(s): `results/moduleB/phenotype_tiers/表型分层_镜像稳定性汇总.csv`, `锚定支持分解_99候选.csv`, `layer1_DE_GSE117525.csv`, `layer1_DE_GSE144304.csv`; generating script **47_moduleB_表型分层镜像稳定性与锚定支持分解_20261015.R** (run 2026-10-15; R 4.6.0, RobustRankAggreg + limma).

Table S25 External-calibration decomposition and recalibration (GSE57218). Calibration was assessed by regressing validation outcomes on the model linear predictor lp (calibration slope) and on lp with an offset (intercept update). Observed: slope = 0.268, intercept update = +5.45. Decomposition: the prevalence shift between layers (55.6% [65/117] training to 82.5% [33/40] validation) implies an intercept shift of Δlogit = 1.33; the fitted update (+5.45) exceeds this by 4.12 logits, i.e. 75.6% of the intercept drift is not explained by prevalence. Consistently, case–control separation of lp shrank from 12.64 (training) to 6.98 (GSE57218), and marker z-score distributions were compressed toward zero in the validation layer (per-marker means and SDs by layer and group in the source CSV)—together indicating cross-platform z-score transport compression rather than a prevalence effect alone. Recalibration: the prespecified minimal intercept-only update (lp' = lp + 5.45) restored calibration-in-the-large (mean prediction 0.582 → 0.825 = observed prevalence), improved the Brier score (0.280 → 0.116), and left discrimination unchanged (AUC = 0.818 before and after); full intercept + slope refitting (lp' = 2.038 + 0.268 × lp; Brier 0.099) is reported only as an in-sample sensitivity, not as a validated model. The panel is therefore advanced strictly as a discrimination-tier prototype requiring cohort-specific recalibration. Source file(s): `results/moduleI/recalibration/校准分解_汇总.csv`, `z分布对比_十标志物_训练层vsGSE57218.csv`, `lp分布_按组对比.csv`, `再校准前后_预测概率.csv`; generating script *48_moduleI_校准失败定量分解与外部再校准_20261015.R* (run 2026-10-15; R 4.6.0; assembly line-identical to script 23, AUC self-check 0.818 ± 0.005 passed).

Table S26 Cell-composition-adjusted sensitivity analysis of the dual-track structure. Motivation: age-related shifts in tissue cellular composition could artefactually generate opposite-direction (mirror) signals. Procedure: for each of the 10 bulk cohorts, per-cohort differential-expression models were refit with principal components of inferred cell-type fractions as covariates (xCell, 67 cell types; ssGSEA, 28 immune/stromal signatures), retaining the smallest PC set explaining ≥ 80% of cell-fraction variance (xCell: k = 8–15 PCs, 80.2–83.0% variance explained; ssGSEA: k = 4–9 PCs, 81.0–82.5%); all cohorts met the residual-df ≥ 4 requirement in both variants. Direct rebuild: adjusted reruns yielded only 17 (xCell) and 23 (ssGSEA) dual-track candidates, with mirror fractions of 41.2% (7/17; one-sided binomial P = 0.834) and 56.5% (13/23; P = 0.339)—nominally nonsignificant but uninformative at these sizes after the > 75% loss of candidates. Direction-retention analysis (informative test): among the 99

frozen candidates, 61 (xCell) and 62 (ssGSEA) retained both-side directions after adjustment; within these retained sets the mirror fraction was 73.8% (45/61; $P = 1.32 \times 10^{-4}$) and 75.8% (47/62; $P = 2.89 \times 10^{-5}$), statistically indistinguishable from the unadjusted 72.7% (χ^2 P = 1.000 and 0.804); 31 mirror genes were retained under both adjustment variants. Attrition was track-symmetric (mirror genes retained 62.5%/65.3% versus concordant 59.3%/55.6%; χ^2 P = 0.950/0.511), arguing against selective cell-composition confounding of the mirror track. Caveats: 38/99 (xCell) and 37/99 (ssGSEA) candidates lost at least one side's direction—including GADD45A on the OA side in both variants—consistent with power loss but not excluding partial composition effects; age itself cannot be adjusted in the aging-proxy cohorts (GSE1428/GSE25941), where age defines the contrast. Source file(s): `results/moduleB/cell_composition_sensitivity/细胞校正_队列协变量汇总.csv`, `细胞校正_镜像稳定性汇总.csv`, `细胞校正_99候选方向保持.csv`, `重大6_摘要.txt`; generating script `49_moduleB_细胞组成校正敏感性_镜像方向重估_20261015.R` (run 2026-10-15; R 4.6.0, xCell + GSVA/ssGSEA + limma + RobustRankAggreg).

Table S27 Instrument-tier and strength disclosure for the 32 primary MR combinations. Under the tiered instrument-construction protocol (Methods), primary parameters ($P < 5 \times 10^{-8}$, $r^2 < 0.001$, 10,000 kb) sufficed for 12/32 combinations; relaxed pruning ($r^2 = 0.01$, 10,000 kb) was required for 12 (the eQTLGen-blood CBR3, GADD45A and HEMK1 pairs); and 8 combinations fell back to a single-SNP Wald ratio (all GTEx-muscle CBR3 and HEMK1 pairs). Instrument strength satisfied $F = (\beta/se)^2 > 10$ in all 32 combinations (per-combination minimum F range 30.2–73.4). Steiger directionality was correct in 32/32 combinations. The six nominally significant candidates span all three tiers: primary (FBLN5–appendicular lean mass, GDE1–low grip strength, RNF14–walking pace), relaxed (CBR3–knee OA, HEMK1–appendicular lean mass eQTLGen), and single-SNP Wald ratio (HEMK1–appendicular lean mass GTEx). Source file: `results/moduleH/disclosure/MR披露_32组合工具层级与F统计.csv`; generating script `50_moduleH_MR披露汇总_工具变量阶梯放宽与F统计_20261015.R` (run 2026-10-15; R 4.6.0).

Table S28 Positive-control MR pipeline validation on 19 genes with published causal evidence. To verify that the MR pipeline can recover established causal signals, the identical pipeline (instrument construction at $P < 5 \times 10^{-8}$ with tiered LD relaxation and single-SNP fallback, harmonisation, IVW/Wald-ratio primary analysis, Steiger directionality, F-statistic

screen, and gene × outcome × eQTL-source family multiplicity control) was rerun on 19 genes with published causal evidence (13 OA-side, 6 sarcopenia-side; Appendix D) against the same four GWAS outcomes and two eQTL sources; colocalization was not run for this control round (the local eQTLGen full-cis file covers only the ten study markers), so coloc-based confirmation was not applicable. 16/19 genes had constructible instruments in ≥ 1 combination (86 primary tests; instrument tiers: primary 28, relaxed 16, single-SNP fallback 42; all F > 10, range 30.6–110.8; Steiger direction correct in 86/86); ITIH1, DNAJB12 and IL12B had no cis-eQTL at $P < 5 \times 10^{-8}$ in either source (structural negative, not a power failure). Under Bonferroni correction across the 86 tests, 13 tests across 7/19 genes (36.8%) remained significant, with recovery sharply outcome-patterned: 5/6 sarcopenia-side genes (HLA-DRA, MAP3K3, MFGE8, COL15A1, AURKA; HP not recovered, best nominal P = 0.088) versus 2/13 OA-side genes (MAPK3, FES); 12/13 significant tests targeted muscle-trait outcomes (appendicular lean mass 8, grip strength 2, walking pace 2) and only 1/13 targeted knee OA (MAPK3; IVW b = −0.090, corrected P = 0.024). MAPK3 showed cross-source concordant effects on lean mass (eQTLGen IVW b = −0.041; GTEx-muscle Wald b = −0.077). The same pipeline that returned no Bonferroni-significant association for the ten study markers thus recovers established signals in 13/86 tests (15.1%), and its sensitivity gap is concentrated on the OA side—quantifying, rather than merely asserting, the eQTL tissue-mismatch limitation discussed in the main text. Source file(s): `results/moduleH/positive_controls/moduleH_MR_summary.csv`, `moduleH_MR_all_methods.csv`, `moduleH_known_causal_overlap.csv`, `moduleH_status.csv`; generating script `51_moduleH_阳性对照_19 已知因果基因 MR 管线复核_20261016_v3.R` (run 2026-10-16; R 4.6.0; TwoSampleMR + ieugwasr/OpenGWAS).

3. TRIPOD+AI (2024) reporting checklist

TRIPOD+AI (2024) checklist

Checklist source. This checklist was transcribed item by item from the officially published TRIPOD+AI version: Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 2024;385:e078378. doi:10.1136/bmj-2023-078378 (also published in *Nat Med* and other journals; indexed by the EQUATOR Network). Item text

follows the official checklist PDF (Version: 11-January-2024, www.tripod-statement.org), comprising **27 main items and 52 sub-items**; the requirement summaries are working paraphrases aligned with the official English wording, which should prevail in the submitted version.

Completion conventions. - Item applicability labels follow the official definitions: D = development studies only; E = evaluation (validation) studies only; D;E = both. This study uses a "**model development plus external evaluation**" design (development of a 10-biomarker logistic diagnostic model plus three-tier external validation in GSE57218 and other datasets), so D, E, and D;E items were all assessed. - Locations refer to sections of the submitted manuscript and to supplementary tables/figures; page and line numbers will be added at proof stage. - Four status levels: **Reported** (location and content complete) / **Partially reported** (main content reported but gaps exist, explained after the status) / **Not applicable** (the study design means the item has no corresponding content; reason given) / **Not reported** (required by the checklist but not addressed in the manuscript; marked honestly without fabricating locations).

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
Title				
1 (D;E)	Title	Identify the study as developing and/or validating a multivariable prediction model, and specify the target population and the outcome to be predicted	Title: subtitle added, ": Development and external validation of a diagnostic biomarker panel for osteoarthritis"	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
Abstract				
2 (D;E)	Abstract	Report per the TRIPOD+AI for Abstracts checklist (13 items): data sources, sample size, outcome, predictors, modelling method, internal/external validation, discrimination and calibration performance measures (with CIs), conclusions, and registration information	Abstract: structured as background–methods–results–conclusions; reports 21 public datasets, four-algorithm screening plus logistic diagnostic model, cartilage primary-validation AUC 0.818 (95% CI 0.614–1.000, DeLong), blood 0.605, seven-classifier external AUC 0.788–0.887, and the positioning of internal evaluation as	**Partially reported** — the abstract does not include training/validation sample sizes, calibration metrics (calibration slope/C-index), data and code availability, or a registration statement

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			reference only	
Introduction				
3a (D;E)	Background: context and rationale	Explain the healthcare context (diagnostic or prognostic) and the rationale for developing or evaluating the model, including reference to existing models	Introduction (osteosarcopenia comorbidity burden, lack of cross-disease diagnostic biomarkers and causal evidence, limitations of prior concordant-intersection studies); prior cross-disease work [6,7] cited in the Introduction; existing OA blood-signature models [37,38] cited in the Discussion	**Reported**
3b (D;E)	Background: target population and intended use	Describe the target population and the intended	Target population (OA cartilage/synovia	**Partially reported** — the model's position

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		use of the model in the care pathway, including intended users (healthcare professionals/patients/public)	l tissue cohorts) and intended-use positioning in the Introduction (deriving diagnostic biomarkers validatable across cohorts; the AUC>0.7 claim restricted to the cartilage layer); "prototype, requiring calibration before entering any clinical context" stated in the Discussion (limitations)	in the clinical pathway and its intended users are not explicitly described; an explicit sentence should be added to the Introduction or Methods: "a research-grade diagnostic stratification tool, not for clinical deployment at this stage, with researchers as intended users"
3c (D;E)	Background: health inequalities	Describe any known health inequalities between sociodemographic groups	Introduction, end of first paragraph: higher OA burden in women,	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			concentration in older adults, and extrapolation limits of evidence dominated by European-ancestry cohorts	
4 (D;E)	Objectives	Specify the study objectives and state whether the study concerns model development, validation, or both	Introduction, final paragraph: development (four-algorithm cross-screening plus logistic modelling) and evaluation (three-tier external validation) pursued together; hypotheses H1–H3 stated explicitly (H2: cross-cohort diagnostic performance of	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			the cartilage primary-validation layer)	
Methods				
5a (D;E)	Data sources	Describe the sources of the development and evaluation datasets separately (cohorts/registries etc.), the rationale for their use, and data representativeness	Methods, “Study design and data sources” (18 bulk + 3 single-cell/single-nucleus GEO datasets, role stratification); training layer of 5 OA cohorts in Methods, “Biomarker screening and diagnostic modelling”; primary validation GSE57218 (RAAK cohort [13]) in Methods, “Study design and data sources” and	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			"External validation and classifier robustness"; representativeness discussed in the Discussion (limitations)	
5b (D;E)	Data dates	Specify the start and end dates of participant data collection and the follow-up cut-off date (if applicable)	GEO retrieval/access date (August 2026) in Methods, "Study design and data sources" and Methods, "Statistical analysis"; enrolment/follow-up dates of participants in each original cohort not itemized	**Partially reported** — original enrolment dates are not obtainable for a secondary analysis; a note should be added that "enrolment dates of each cohort are as reported in the original publications"
6a (D;E)	Setting	Describe key elements of the study setting	Tissue sources, platforms, and roles of each	**Partially reported** — centre/setting

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		(primary/secondary care/general population) and the number and location of centres	dataset in Methods, “Study design and data sources” and Table S16; RAAK cohort citation [13]; number of centres and geographic locations of each cohort not itemized	fields should be added to Table S16 (marked as unavailable where not obtainable)
6b (D;E)	Eligibility criteria	Describe the inclusion and exclusion criteria for study participants	Sample-level inclusion/exclusion at the secondary-analysis level is explicit (Methods, “Study design and data sources”: rheumatoid arthritis samples excluded, baseline samples	**Partially reported** — a pointer sentence should be added: “participant-level criteria are as reported in the original studies”

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			only, platform restrictions, restricted use of pooled samples, etc.); participant-level diagnostic/enrolment criteria of the original cohorts not restated	
6c (D;E)	Treatment information	Detail treatments received by participants and how they were handled in model development/evaluation (if relevant)	Methods, end of the eligibility key-points paragraph: public datasets did not provide per-sample treatment/medication information; treatment exposure was not included in modelling or evaluation	**Reported**
7 (D;E)	Data preparation	Describe data	Methods,	**Partially**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		preprocessing and quality checks, and whether preprocessing was consistent across sociodemographic subgroups	"Preprocessing, differential expression, and dual-track construction": GEOquery retrieval, RMA + quantile normalization + log2, probe re-annotation, ComBat batch correction with before/after PCA checks, duplicateCorrelation handling of within-donor correlation; preprocessing applied uniformly to all samples within each dataset	**reported** — consistency across subgroups not explicitly declared (public metadata mostly lack sociodemographic fields); a statement should be added that "the preprocessing pipeline was identical for all samples and did not vary by demographic characteristics"
8a (D;E)	Outcome definition	Clearly define the outcome to be predicted and	The outcome is OA disease status	**Partially reported** — a standalone

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		the time window, how and when it was assessed, the rationale for its choice, and whether outcome assessment was consistent across subgroups	(diagnostic model), inferable from Methods, “Preprocessing, differential expression, and dual-track construction” (case–control differential analysis), Methods, “Biomarker screening and diagnostic modelling” (training set 65 cases/52 controls), and Methods, “External validation and classifier robustness” (case/control counts per layer); no standalone	“outcome definition” sentence should be added to the Methods: the outcome is each original cohort’s diagnostic label (OA vs control); diagnostic prediction has no time window; label adjudication criteria are as per the original studies

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			outcome definition is given	
8b (D;E)	Outcome assessors	If outcome assessment requires subjective interpretation, describe the qualifications and demographic characteristics of the assessors	No new outcome assessment in this study; existing public dataset labels were used directly	**Not applicable** — pure public-data secondary analysis with no study-level outcome assessors
8c (D;E)	Outcome blinding	Report any blinding of outcome assessment	No prospective data collection; outcome labels pre-existed the analysis	**Not applicable** — no blinding of outcome assessment is possible (marked honestly)
9a (D)	Initial predictors	Describe the basis for selecting initial predictors (literature/existing models/all available) and	Methods, “Biomarker screening and diagnostic modelling”: the feature space comprised the 99	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		any pre-modelling screening	dual-track genes ∩ genes detectable in each training cohort ∩ genes detectable on the GPL6947 platform (reduced to 84; platform information available a priori); restated in Results, “Four-algorithm cross-screening identifies a 10-biomarker panel”	
9b (D;E)	Predictor definition	Clearly define all predictors, how and when they were measured (and any blinding to the outcome or other predictors)	The 10-biomarker list in Results, “Four-algorithm cross-screening identifies a 10-biomarker panel” (NELL1,	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			STEAP1, GADD45A, FBLN5, HEMK1, GDE1, RNF14, BNIP3, EGFR, CBR3); measurement by microarray/RNA-seq expression, per-cohort z-scores (Methods, “Biomarker screening and diagnostic modelling”); objective instrument measurement, no blinding required	
9c (D;E)	Predictor assessors	If predictor measurement requires subjective interpretation, describe the qualifications	Predictors are transcriptomic instrument measurements	**Not applicable** — no subjective interpretation involved

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		and demographic characteristics of the assessors		
10 (D;E)	Sample size	Explain how the sample size was determined (development and evaluation separately) and justify its adequacy, including any sample-size calculation	Per-layer sample sizes reported honestly: training 117 (65/52) (Methods, “Biomarker screening and diagnostic modelling”; Results, “Four-algorithm cross-screening identifies a 10-biomarker panel”); GSE57218 40 cases (33 OA/7 controls) (Methods, “External validation and classifier	**Partially reported** — no formal sample-size estimation or adequacy justification (e.g., Riley et al. pmsampsize criteria); a qualitative justification or a note on the effect of small samples on CI width should be added

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			robustness"; Results, "Diagnostic model construction, robustness, and three-tier external validation"); GSE168505 n=7, down-weighted (Results, "Diagnostic model construction, robustness, and three-tier external validation"; Discussion, limitations); GSE48556 139 cases (Methods, "External validation and classifier	

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			robustness”)	
11 (D;E)	Missing data	Describe how missing data were handled, and give reasons for discarding any data	Dataset/feature-level missing-data rules are explicit: GSE206848 (marker platform coverage 0/10) not subjected to validation per the prespecified rule (Methods, “External validation and classifier robustness”; Results, “Diagnostic model construction, robustness, and three-tier external validation”), GSE136344 (empty intersection)	**Partially reported** — handling of within-sample missing expression values (imputation/exclusion) not stated; one sentence should be added

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			(Results, “Diagnostic model construction, robustness, and three-tier external validation”), GSE12021 (restricted to the GPL96 platform; 19 samples entered the merged training matrix) (Methods, “Study design and data sources”), GSE129147 (removed from the main analysis) (Methods, “Study design and data sources”)	

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
12a (D)	Data use and partitioning	Describe how data were used in the analysis (development vs performance evaluation), whether any partitioning was performed, and sample-size considerations	Methods, “Biomarker screening and diagnostic modelling” and “External validation and classifier robustness”: 5 OA cohorts z-score-merged into the training layer; the “cartilage primary validation—blood exploration—muscle-proxy exploration” three-tier external validation architecture was specified before analysis; internal evaluation was repeated 10×10-	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			fold CV	
12b (D)	Predictor handling	Describe how predictors were handled in the analysis (functional form, rescaling, transformation, standardization)	Per-cohort z-score standardization followed by merging (Methods, “Biomarker screening and diagnostic modelling”); log2 transformation (Methods, “Preprocessing, differential expression, and dual-track construction”); logistic model fitted on z-score expression (Methods, “Biomarker screening and diagnostic modelling”)	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
12c (D)	Model type and construction	Specify the model type and rationale, all model-building steps (including hyperparameter tuning), and internal validation methods (listed per modelling method)	Methods, “Biomarker screening and diagnostic modelling”: LASSO (λ.1se criterion), SVM-RFE, RF, and XGBoost applied as four independent screens, with LASSO∩SVM-RFE as the primary route and RF/XGBoost top10 voting as auxiliary; the final model was logistic regression; internal validation was repeated 10×10-fold CV (B=2000 bootstrap CI);	**Reported** — per-algorithm hyperparameter details are reproducible via the public code (Methods, “Statistical analysis”; Declarations)

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			selection leakage arising from feature selection performed outside CV was honestly declared; nomogram bootstrap (B=200) optimism correction; seven classifiers under the caret framework with 3-times-repeated 10-fold CV, per-method fixed seeds 20260914–20260921, and xgbTree dropped for interface incompatibility with an explanatory note (Methods, “External	

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			validation and classifier robustness")	
12d (D;E)	Cluster heterogeneity	Describe and quantify the handling of heterogeneity in model parameters and performance across clusters (hospitals/countries)	Per-cohort z-score plus RRA cross-cohort integration (Methods, "Preprocessing, differential expression, and dual-track construction"); single-gene cross-cohort direction heterogeneity discussed (Results, "Four-algorithm cross-screening identifies a 10-biomarker panel"); direction-consistency summary for the	**Partially reported** — cross-cluster heterogeneity in model performance was not formally quantified under the TRIPOD-Cluster framework; an explanatory note should be added, or TRIPOD-Cluster cited with a per-cohort performance summary table (partly covered by Tables S11/S12)

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			muscle-proxy layer (Results, “Diagnostic model construction, robustness, and three-tier external validation”)	
12e (D;E)	Performance measures	Specify all performance measures and plots (discrimination, calibration, clinical utility) with rationale, and methods for comparing multiple models	“Three-dimensional model evaluation (TRIPOD+AI)” in Methods, “Biomarker screening and diagnostic modelling”: ROC-AUC + DeLong 95% CI, calibration curves (including the Hosmer–Lemeshow test), and DCA; C-	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			index and calibration slope additionally reported (Methods, “Biomarker screening and diagnostic modelling”), with apparent vs optimism-corrected metrics (Results, “Diagnostic model construction, robustness, and three-tier external validation”); seven-classifier external AUC comparison under the same metrics (Methods, “External validation and	

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			classifier robustness"; Results, "Diagnostic model construction, robustness, and three-tier external validation")	
12f (E)	Model updating	Describe any model updating arising from the evaluation (e.g., recalibration), overall or for specific subgroups/settings	No model updating was performed; external calibration slope 0.268 and intercept 5.45 were reported diagnostically only (Results, "Diagnostic model construction, robustness, and three-tier external validation");	**Not applicable** — no model updating (marked honestly)

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			recalibration is listed as future work (Discussion, limitations)	
12g (E)	Prediction calculation	Specify how model predictions were calculated during evaluation (formula, code, object, API)	rms::lrm fit (11 coefficients including the intercept; Methods, “Biomarker screening and diagnostic modelling”; Results, “Diagnostic model construction, robustness, and three-tier external validation”); external layers received z-score inputs under the same convention as training, with	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			the ROC direction explicitly fixed (Methods, “Biomarker screening and diagnostic modelling”); all analysis code is public on GitHub (Methods, “Statistical analysis”; Declarations)	
13 (D;E)	Class imbalance	If class-imbalance methods were used, explain why, how, and any subsequent recalibration of the model/predictions	No oversampling, class weighting, or any other class-imbalance method was used (training set 65/52, mildly imbalanced; Methods, “Biomarker screening and	**Not applicable** — no such methods were used; an explicit sentence stating “no class-imbalance handling was applied” should be added to the main text

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			diagnostic modelling")	
14 (D;E)	Fairness	Describe any methods used to address model fairness, with rationale	No dedicated fairness methods were implemented; extrapolation limits arising from the blood cohort's sex composition (129 female/10 male) are declared in the limitations (Methods, "External validation and classifier robustness"; Discussion, limitations)	**Partially reported** — an explicit sentence should be added stating that "no dedicated fairness methods were used, and why (sociodemographic fields are incomplete in public metadata)"
15 (D)	Model output	Specify the model output type (probability/classification) and,	The output is a predicted probability (out-of-bag probabilities	**Partially reported** — no fixed classification threshold was

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		for any classification, the details and rationale of threshold determination	averaged; Methods, “Biomarker screening and diagnostic modelling”; classProbs=TRUE; Methods, “External validation and classifier robustness”); evaluation was conducted across full-threshold ROC/calibration/DCA metrics	specified; an explicit sentence should be added: “this study sets no clinical classification threshold; the output is a continuous probability”
16 (D;E)	Training–evaluation differences	Identify any differences between development and evaluation data in healthcare setting, eligibility criteria, outcome, and	Tissue and platform differences of the three-tier architecture were prespecified (Methods, “External validation and	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		predictors	classifier robustness": cartilage primary validation / PBMC blood / muscle proxy); the blood layer's sib-pair family structure was handled by switching to unpaired analysis (Methods, "External validation and classifier robustness"); case-by-case handling of the divergence between GDE1 group-mean and rank-based metrics (Results, "Diagnostic model construction,	

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			robustness, and three-tier external validation")	
17 (D;E)	Ethical approval	Name the institutional review board/ethics committee that approved the study, and describe informed consent or its waiver	Declarations: this study is a secondary analysis of publicly available de-identified data, involved no new participant recruitment, and required no additional ethics approval; ethics approvals of the original datasets are as stated in their publications	**Reported**
Open Science				
18a (D;E)	Funding	Give the source of funding and the role of the funder in the	Declarations: this study received no	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		study	specific funding	
18b (D;E)	Conflicts of interest	Declare conflicts of interest and financial disclosures for all authors	Declarations: all authors declare no conflicts of interest	**Reported**
18c (D;E)	Protocol	Indicate where the study protocol can be accessed, or state that no protocol was prepared	Declarations, “Study protocol, registration, and patient involvement” paragraph: no separately registered protocol document was prepared; the analytic key points were executed according to a prespecified internal plan and are reported in the Methods	**Reported**
18d (D;E)	Registration	Provide registration	Same Declarations	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		information (registry name and registration number), or state that the study was not registered	paragraph: this study was not registered	
18e (D;E)	Data sharing	Provide details of the availability of the study data	Data availability statement (Declarations): all 21 datasets are available from GEO (accessions and grouping in Table S16; access date 2026-08); GWAS/eQTL/druggable-genome list sources are complete; intermediate statistics are in Tables S1–S28	**Reported**
18f (D;E)	Code sharing	Provide details of the	Methods, “Statistical	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		availability of the analysis code (data cleaning, feature engineering, modelling, evaluation)	analysis"; Declarations: all code (including per-module explicitly fixed random seeds) is hosted on GitHub (https://github.com/Chiaki74/OA-sarcopenia-dual-track); the CIBERSORT source code and LM22 matrix are not deposited owing to Stanford licence restrictions, with the access route noted	
Patient and public involvement				
19 (D;E)	Patient and public involvement	Provide details of any PPI in study design,	Same Declarations paragraph:	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
	(PPI)	conduct, reporting, interpretation, or dissemination, or state that there was none	public-data secondary analysis; no patient and public involvement	
Results				
20a (D;E)	Participant flow	Describe the flow of participants through the study (numbers with/without the outcome, summary of follow-up time), preferably with a flow diagram	Fig 1, overall flow diagram (integrative analysis design of 21 datasets); dataset-level screening flow in Methods, “Study design and data sources” and Table S16; per-layer sample sizes and outcome counts in Results, “Identification of dual-track shared genes and the dominance	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			of mirror dysregulation", "Four-algorithm cross-screening identifies a 10-biomarker panel", and "Diagnostic model construction, robustness, and three-tier external validation"	
20b (D;E)	Participant characteristics	Report participant characteristics overall and per data source/setting (key dates, demographics and key predictors, treatments, sample size, number of	Table S16 is the dataset-level master table (accession/tissue /platform/actual sample size/role/three-class phenotype); GSE48556 sex composition 129 female/10 male (Methods,	**Partially reported** — individual-level fields are incomplete in public metadata; available cohort demographic information should be tabulated with unavailable items noted one

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		events, follow-up, amount of missing data), preferably in a table; report differences across demographic groups	"External validation and classifier robustness"; Discussion, limitations); a per-cohort summary of individual-level demographics such as age and sex is missing	by one
20c (E)	Development–evaluation distribution comparison	For model evaluation, present a comparison of the distributions of important predictors (demographics, predictors, outcome) in the evaluation data versus the development data	Outcome prevalence compared: GSE57218 82.5% vs training layer 65/52 (Results, "Diagnostic model construction, robustness, and three-tier external validation"; Methods,	**Partially reported** — a distribution comparison of the 10-biomarker z-scores between the training layer and GSE57218 should be added (may go into a supplementary table)

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			"Biomarker screening and diagnostic modelling"); a training-vs-validation distribution table for the predictors (10-biomarker expression) was not provided	
21 (D;E)	Sample size per analysis	Specify the number of participants and outcome events in each analysis (model development, hyperparameter tuning, model evaluation)	Training layer 117 (65/52) (Results, "Four-algorithm cross-screening identifies a 10-biomarker panel"); internal CV and tuning based on the same training set (Methods, "Biomarker screening and diagnostic modelling";	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			Results, “Diagnostic model construction, robustness, and three-tier external validation”); per-layer external n and event counts: GSE57218 33/7, GSE168505 4/3, GSE48556 106/33 (Methods, “External validation and classifier robustness”; Results, “Diagnostic model construction, robustness, and three-tier external validation”);	

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			muscle-proxy layer (Results, “Diagnostic model construction, robustness, and three-tier external validation”)	
22 (D)	Full model specification	Provide full details of the prediction model (formula/code/object/API) to support prediction in new individuals, third-party evaluation, and implementation, including any access/reuse restrictions	Full coefficient table (10 coefficients plus intercept, ORs and Wald 95% CIs) in the Table S13 supplement; nomogram in Fig 3f; implementation code publicly available (Methods, “Statistical analysis”; Declarations)	**Reported** (Table S13 supplement)
23a (D;E)	Model performance	Report performance	External AUC 0.818 (95% CI	**Partially reported** — the

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		estimates with confidence intervals, including in key subgroups (e.g., sociodemographic); plots should preferably aid presentation	0.614–1.000, DeLong; Results, “Diagnostic model construction, robustness, and three-tier external validation”; Fig 4a); internal CV AUC 0.959 (0.92–0.99) (Results, “Four-algorithm cross-screening identifies a 10-biomarker panel”); bootstrap-corrected C-index 0.97 and corrected calibration slope 0.394 (Results, “Diagnostic model construction,	main metrics (discrimination/calibration/clinical utility + CIs + plots) are complete; performance in key sociodemographic subgroups was not evaluated, with only a qualitative statement on the blood cohort’s sex composition (Discussion, limitations)

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			robustness, and three-tier external validation"; Table S13); external calibration slope 0.268/intercept 5.45 and calibration curve (Results, "Diagnostic model construction, robustness, and three-tier external validation"; Fig 4b); DCA (Fig 4c); seven-classifier external AUC 0.788–0.887 (Results, "Diagnostic model construction, robustness, and	

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			three-tier external validation"; Table S12); blood 0.605 (0.499–0.713; Results, "Diagnostic model construction, robustness, and three-tier external validation")	
23b (D;E)	Performance heterogeneity across clusters	If examined, report heterogeneity of model performance across clusters	Performance reported per cohort under the prespecified three-tier architecture (Results, "Diagnostic model construction, robustness, and three-tier external	**Reported** — no formal heterogeneity statistic was computed; results are presented per the prespecified stratification

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			validation"); single-gene cross-cohort direction heterogeneity discussed (Results, "Four-algorithm cross-screening identifies a 10-biomarker panel"); the muscle-proxy layer did not meet its prespecified ≥ 2-cohort acceptance criterion (2/10; 45% concordance, below the 50% chance expectation) and provided no supportive evidence (Results,	

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			"Diagnostic model construction, robustness, and three-tier external validation")	
24 (E)	Model updating results	Report the results of any model updating, including the updated model and its subsequent performance	No model updating or recalibration was performed	**Not applicable** — no updating; recalibration is listed as future work (Discussion, limitations)
Discussion				
25 (D;E)	Interpretation	Give an overall interpretation of the main results in the context of the study objectives and previous research, including fairness issues	Discussion: the dominance of mirror dysregulation, cartilage-layer panel performance versus the blood-layer tissue gap, systematic comparison with	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			Yang et al. [6]/Wang et al. [7] and trait-level genetic studies [9,10], and an honest interpretation of overfitting; the fairness-related (sex composition) statement is in the limitations	
26 (D;E)	Limitations	Discuss study limitations (sample representativeness, sample size, overfitting, missing data, etc.) and their implications for bias, statistical uncertainty, and generalisability	Limitations (Discussion): the phenotype-proxy strategy, blood-layer downgrade and the tissue gap, sample size and statistical power (including the overfitting evidence chain of calibration slope 0.394→external	**Reported**

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
			0.268), MR population and tissue representativeness limitations, and compensation strategies for a purely dry-lab design	
27a (D)	Implementation: input data quality	Describe how poor-quality or unavailable input data (predictor values) should be assessed and handled when the model is implemented	The model is explicitly positioned as a prototype at this stage and does not enter any clinical-use context (Discussion, limitations); input-data quality control at implementation was not discussed	**Not applicable** — no clinical deployment is claimed; the model is positioned as a research-grade diagnostic stratification tool (marked honestly)
27b (D)	Implementation: user interaction	Specify whether users need to be	As above; user interaction and	**Not applicable** — no clinical

Item	Topic	Requirement (summary of official wording)	Location in this manuscript	Status
		involved in handling input data or using the model, and the level of expertise required	qualification requirements were not addressed	deployment is claimed (marked honestly)
27c (D;E)	Further research	Discuss future research steps, focusing on the model's applicability and generalisability	Discussion: development of a "blood-adapted" biomarker panel; recalibration in larger cohorts with representative prevalence; pQTL colocalization and Phe-MR expansion plans; spatial transcriptomics	**Reported**

Gap summary

Not-reported items (0; all 6 originally open items were resolved in the submitted manuscript)

Item	Original gap	Resolution
1 (Title)	The title did not identify the study as prediction-model development and evaluation, nor the target population and outcome	Subtitle added to the title: ": Development and external validation of a diagnostic biomarker panel for osteoarthritis"
3c (Health inequalities)	Known health inequalities between sociodemographic groups not described	A sentence on demographic inequalities in disease burden added at the end of the Introduction's first paragraph (GBD and other citations added in the English version)
6c (Treatment information)	Participant treatment information not addressed	End of the Methods eligibility key-points paragraph states that treatment information was unavailable and was not included in modelling or evaluation
18c (Protocol)	Protocol availability not stated	A new "Study protocol, registration, and patient involvement" paragraph added to the Declarations
18d (Registration)	Registration status not stated	The same paragraph states "this study was not registered"
19 (PPI)	Patient and public involvement not stated	The same paragraph states "no patient and public

Item	Original gap	Resolution
		involvement”

Partially reported items (15) and completion recommendations

Item	Gap	Completion recommendation
2 (Abstract)	Missing sample sizes, calibration metrics, data/code availability, and registration statement	Complete per TRIPOD+AI for Abstracts (13 items): training/validation sample sizes, corrected calibration slope, GitHub repository, and a “not registered” sentence
3b (Target population and intended use)	Position in the clinical pathway and intended users not made explicit	Add to the Introduction/Methods: “a research-grade diagnostic stratification tool, with researchers as intended users, not for clinical deployment at this stage”
5b (Data dates)	Enrolment/follow-up dates of the original cohorts not listed	Add “enrolment dates of each cohort are as reported in the original publications”; a column may be added to Table S16 (marked as unavailable where not obtainable)
6a (Setting)	Number of centres and locations of each cohort not itemized	Add setting/centre fields to Table S16 (marked as unavailable where not obtainable)

Item	Gap	Completion recommendation
6b (Eligibility criteria)	Participant-level diagnostic/enrolment criteria of the original cohorts not restated	Add the pointer sentence "participant-level criteria are as reported in the original studies"
7 (Data preparation)	Consistency of preprocessing across subgroups not declared	Add "the preprocessing pipeline was applied identically to all samples within each dataset and did not vary by demographic characteristics"
8a (Outcome definition)	No standalone outcome definition	Add to the Methods: "the outcome is each original cohort's OA diagnostic label (affected vs control); diagnostic prediction has no time window; adjudication criteria are as per the original studies"
10 (Sample size)	No formal sample-size estimation/adequacy justification	Add a qualitative justification or a note on the effect of small samples on CI width (e.g., citing the Riley et al. sample-size criteria as a comparative discussion)
11 (Missing data)	Handling of within-sample missing expression values not stated	Add one sentence on the missing-value handling convention (or state that no missing values remained after

Item	Gap	Completion recommendation
		preprocessing)
12d (Cluster heterogeneity)	Cross-cluster heterogeneity not formally quantified per TRIPOD-Cluster	Add an explanatory sentence (multi-cohort sources, per-cohort z-score and RRA handling logic), or cite TRIPOD-Cluster and provide a per-cohort performance summary
14 (Fairness)	No fairness methods used and not stated	Add an explicit sentence: “no dedicated fairness methods were used, and why (sociodemographic fields are incomplete in public metadata)”; retain the sex-composition statement
15 (Model output)	No classification threshold specified	Add “the output is a continuous predicted probability; this study sets no clinical classification threshold”
20b (Participant characteristics)	Summary of individual-level demographics missing	Tabulate available cohort demographic information, noting unavailable items one by one
20c (Development–evaluation distribution comparison)	Predictor distribution comparison table missing	Add a distribution comparison of the 10-biomarker z-scores between the training layer and

Item	Gap	Completion recommendation
		GSE57218 (may go into a supplementary table)
23a (Model performance)	Subgroup performance by sociodemographics not evaluated	Add one sentence: “subgroup performance evaluation was not performed owing to public metadata limitations”; if sex fields are available for platforms such as GSE48556, exploratory subgroup AUCs may be computed

Editorial notes (not item gaps)

1. Main-text figure cross-references were unified to Fig 3f, and a recheck found no residual old figure numbers.
2. Although item 13 (class imbalance) and items 12f/24 (model updating) are marked “Not applicable”, an explicit sentence should be added to the main text/Declarations for each, to satisfy the TRIPOD+AI requirement that non-applicable items should also be explicitly stated.
3. The completeness of Abstract item 2 should be checked separately against TRIPOD+AI for Abstracts (13-item sub-checklist).

Completeness statistics

- 27 main items and 52 sub-items (this study is a model development plus external evaluation design; all 52 items were assessed).
- **Reported: 29 items (55.8%)**; **Partially reported: 15 items (28.8%)**; **Not applicable: 8 items (15.4%)**; **Not reported: 0 items (0%)**.

- **Not-applicable items: 8b, 8c, 9c, 12f, 13, 24, 27a, 27b (reasons given in the table).**

4. STROBE-MR (2021) checklist and Appendix D

Locations refer to sections of the submitted manuscript and to supplementary tables/figures; page and line numbers will be added at proof stage.

Part 1. STROBE-MR checklist (MR and colocalization)

Checklist source (official published items): Skrivankova VW, Richmond RC, Woolf BAR, et al. Strengthening the Reporting of Observational Studies in Epidemiology Using Mendelian Randomization: The STROBE-MR Statement. *JAMA*. 2021;326(16):1614–1621. doi:10.1001/jama.2021.18236 (20 main items, 30 subitems, 6 sections); accompanying explanation and elaboration (E&E): Skrivankova VW, et al. *BMJ*. 2021;375:n2233. doi:10.1136/bmj.n2233. Item topics and requirement summaries are paraphrased from the official wording of JAMA Table 2 / BMJ E&E Table 3.

Scope: The MR component of this manuscript is a two-sample, summary-data, cis-eQTL → GWAS design (TwoSampleMR + coloc colocalization).

Locations refer to sections of the submitted manuscript and to supplementary tables/figures; page and line numbers will be added at proof stage.

Status definitions: Reported (the main text or explicitly indicated supplementary material fully covers the item requirement); Partially reported (only some elements of the item are covered); Not applicable (the item does not apply to this summary-data design, with reason given); Not reported (should have been reported but was not found, constituting a gap).

Item	Topic	Requirement (summary)	Location in this manuscript	Status
1	Title and	Indicate MR as	Abstract	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
	abstract	the study design in the title and/or abstract if MR is the main purpose of the study	(Methods paragraph) states "two-sample Mendelian randomization (MR) and Bayesian colocalization"; Keywords include "Mendelian randomization"; the Title does not contain "MR" (MR is one component of the integrative analysis pipeline rather than the main objective, consistent with the E&E exception note)	
2	Background	Explain the scientific background and rationale; the exposure and	Introduction; rationale that single-disease MR studies exist while cross-	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
		outcome; plausibility of a causal relationship; justify why MR is an appropriate method	disease shared-gene causal testing remains a gap; H3 hypothesis	
3	Objectives	State specific objectives and pre-specified causal hypotheses; state that MR is a method that estimates causal effects under specific assumptions	Study objectives and pre-specified hypotheses H1–H3 in the Introduction (H3 is the causal candidate hypothesis); statement that MR relies on three core assumptions (Methods, “Mendelian randomization and colocalization”)	Reported
4	Study design and data sources	Present key elements of the study design	Methods, “Study design and data sources”; MR	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
		early in the article; consider a table listing data sources for all phases	data sources: four outcome GWAS IDs, eQTLGen 2019-12-11 release, GTEx v8 skeletal muscle, OpenGWAS retrieval with access dates; 80 candidate exposure×outcome×source combinations; GEO dataset overview in Table S16; no dedicated MR data-source table—enumeration in the main text instead	
4a	Setting	Describe the study design, underlying population, location, and relevant dates	eQTL sample sizes and versions and GWAS access dates (Methods, “Study design	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			and data sources"); predominantly European-ancestry samples (Methods, "Mendelian randomization and colocalization"); recruitment periods and related details per the original publications (refs [14–16])	
4b	Participants	Eligibility criteria, sources and methods of selection; sample size; whether any a priori power/sample-size calculations were carried out	eQTLGen n=31,684 and GTEx v8 skeletal muscle n=706 (Methods); outcome GWAS sample sizes not listed in the main text; no a priori power/sample-	Partially reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			size calculation reported (power discussed in the Discussion, limitations)	
4c	Genetic variants	Describe the measurement, quality control, and selection of genetic variants	Tiered instrumental-variable filtering: $P<5\times10^{-8}$, 1000 Genomes European panel LD pruning ($r^2<0.001$, 10,000 kb), stepwise relaxation when fewer than 3 SNPs are available, Wald ratio for single-SNP instruments, F>10 (Methods, “Mendelian randomization and colocalization”);	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			cis definition; harmonization with action=2 and conservative inference/removal of palindromic SNPs	
4d	Assessment and diagnostic criteria	For each exposure/outcome/relevant variable, describe methods of assessment and diagnostic criteria for diseases	Outcome names and GWAS IDs (Methods, “Study design and data sources”); the specific measurement/diagnostic criteria for appendicular lean mass, low grip strength, walking pace, and knee OA are not elaborated in the main text (traceable to the original publications via the GWAS IDs)	Partially reported
4e	Ethics approval	Provide details	“Ethics approval	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
	and informed consent	of ethics approval and participant informed consent (if relevant)	and consent to participate" statement (Declarations): secondary analysis of publicly available de-identified data requiring no additional ethics approval; original studies are governed by their published statements	
5	Assumptions	Explicitly state the three core IV assumptions for the main analysis (relevance, independence, exclusion restriction) and assumptions for additional/sensitivity analyses	"Core assumptions" paragraph (Methods, "Mendelian randomization and colocalization"): (i) relevance ($P<5\times10^{-8}$, $F>10$), (ii) independence	Partially reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			(European ancestry, principal-component adjustment; residual stratification addressed in the limitations), (iii) exclusion restriction (MR-Egger intercept, MR-PRESSO); additional assumptions of sensitivity analyses (e.g. InSIDE for MR-Egger, majority-valid for the weighted median) not explicitly stated one by one	
6	Statistical methods (main analysis)	Describe the statistical methods and statistics used	Methods, “Mendelian randomization and	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			colocalization" (primary and sensitivity analyses; multiple testing and colocalization); R 4.6.0 and package-version summary in Table S10 (Methods, "Statistical analysis")	
6a	Quantitative variables	Describe how quantitative variables were handled in the analyses (scale, units, model)	eQTL effect scales and outcome units not stated in the main text; individual b values given (HEMK1, two sources: b=+0.0303/+0.0301; Results, "MR, colocalization, and	Partially reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			druggability")	
6b	Handling of genetic variants	Describe how genetic variants were handled in the analyses and, if applicable, how their weights were selected	cis instrument construction and LD pruning (Methods); weights taken from the exposure eQTL summary data; harmonization rules (Methods)	Reported
6c	MR estimators	Describe the MR estimator (e.g. 2SLS, Wald ratio) and related statistics; for two-sample MR, state whether the same covariate set was used in both samples	IVW with multiplicative random effects as the primary analysis, MR-Egger and weighted median as secondary, Wald ratio for single-SNP instruments (Methods); TwoSampleMR (ref [16]); under the summary-data design, covariate	Partially reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			adjustment in the two samples follows the original GWAS; consistency of the covariate sets is not stated in the main text	
6d	Missing data	Explain how missing data were addressed	Two-sample summary-data MR; no individual-level missing-data issue; handling of palindromic/ambiguous SNPs described (Methods)	Not applicable
6e	Multiple testing	Explain how multiple testing was addressed (if applicable)	Bonferroni as primary, BH as sensitivity; the testing family comprises the 32 tests with constructible cis instruments among the 80	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			candidate combinations (Methods; Results; Fig 5 legend)	
7	Assessment of assumptions (methods)	Describe any methods or prior knowledge used to assess the assumptions or justify their validity	F>10 instrument strength (Methods); Cochran's Q, MR-Egger intercept, MR-PRESSO, Steiger directionality (Methods); colocalization self-falsification framework (Methods)	Reported
8	Sensitivity and additional analyses (methods)	Describe sensitivity/additional analyses (comparison of estimates across methods, independent replication, bias-analysis	Q/Egger intercept/PRESSO/Steiger (Methods); BH sensitivity (Methods); coloc dual threshold with PPH4>0.8 as the primary	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
		techniques, instrument validation, simulations, etc.)	judgment threshold and PPH4>0.9 co-reported, aligned with prior standards (Methods); coloc-SuSiE fine-mapping sensitivity (nominally significant combinations among the 24 combinations; L1/L2/L3 shrinkage-LD tiers)	
9	Software and pre-registration	See 9a/9b	—	See subitems
9a	Statistical software	Name the statistical software and package(s), including versions and settings used	R 4.6.0, package versions summarized in Table S10, random seeds explicitly fixed (Methods, “Statistical	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			analysis"); TwoSampleMR [16], coloc [35,36]; code hosted on GitHub (Methods; Data availability)	
9b	Pre-registration	State whether the study protocol and details were pre-registered, and when and where	Declarations ("Study protocol, registration, and patient involvement"): "No separately registered study protocol document was prepared; key analytic steps followed a prespecified internal analysis plan and are reported in the Methods. This study was not registered";	**Reported** — verified 2026-10-08: the statement is present in the Declarations section; this row reflected a checklist update omission rather than a content gap

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			multiple main-text statements that analyses were "prespecified in advance" (Introduction; Methods; Results)	
10	Descriptive data	See 10a–10d	—	See subitems
10a	Numbers at each stage	Report the numbers of individuals at each stage and reasons for exclusion; consider a flow diagram	MR is a summary-data design with no individual-level inclusion/exclusion; sample sizes of the two eQTL sources (Methods); GEO cohort sample sizes and exclusions (Methods, "Study design and data sources"; Table S16); outcome GWAS sample	Partially reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			sizes not listed; Fig 1 is the overall flow diagram	
10b	Summary statistics	Report summary statistics for phenotypic exposure(s), outcome(s), and other relevant variables (means, SDs, proportions)	Summary-data design: exposures are eQTL effects and outcomes are GWAS summary statistics; phenotype-level means/SDs are as per the original GWAS publications and are not listed in the main text	Not applicable
10c	Heterogeneity between data sources	If the data sources include meta-analyses of previous studies, provide assessments of heterogeneity across these studies	Methods, “Study design and data sources” (“GWAS and QTL resources” paragraph): the four outcome GWAS summary	**Reported** — completed in the main text and verified 2026-10-08

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			datasets derive from large biobank or consortium studies, partly pooled multi-cohort meta-analyses; between-source-study heterogeneity statistics (e.g. I^2) are as per the original publications and were not re-assessed here; at the analysis level, between-instrument heterogeneity was assessed with Cochran's Q (Table S6)	
10d	Two-sample MR specific	(1) Justify the similarity of the variant–exposure associations	Similarity: both exposure and outcome GWAS are	Partially reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
		between the exposure and outcome samples; (2) provide the number of individuals overlapping between the two samples	predominantly European-ancestry with principal-component adjustment (Methods); the number of overlapping individuals was not quantified	
11	Main results	See 11a–11d	—	See subitems
11a	Variant associations	Report variant–exposure and variant–outcome associations on an interpretable scale	Combination-level instrument counts and P values (Results; Table S17); per-SNP variant–exposure/variant–outcome associations not presented in the main text (whether they are fully listed in the supplementary tables cannot be	Partially reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			confirmed from this manuscript)	
11b	MR estimates	Report MR estimates and measures of uncertainty on an interpretable scale (e.g. OR/RR per SD)	IVW/Wald P values and instrument counts for the 6 nominal candidates (Results); HEMK1 b values from two sources (Results); 95% CIs not given individually in the main text	Partially reported
11c	Absolute-risk translation	If relevant, translate relative-risk estimates into absolute risk for a meaningful time period	Not performed; the primary results were null after correction and the candidates are nominally significant leads, so absolute-risk translation is not applicable	Not applicable
11d	Visualization of	Consider plots to	Fig 5a MR forest	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
	results	visualize results (forest plot, scatterplot of variant–outcome vs variant–exposure associations, etc.)	plot, 5b colocalization PPH4/PPH3 summary, 5c abf–SuSiE PPH4 concordance scatter, 5d locuszoom regional LD plots	
12	Assessment of assumptions (results)	See 12a/12b	—	See subitems
12a	Results of assumption-validity assessment	Report the assessment of the validity of the assumptions	Steiger directionality passed 32/32 (Results); sensitivity analyses revealed no bias sufficient to overturn the conclusions	Reported
12b	Additional statistics	Report any additional statistics (e.g. between-variant	Cochran’s Q, MR-Egger intercept, and MR-PRESSO results in Table	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
		heterogeneity I^2/Q statistic or E-value)	S6 (Fig 5a legend)	
13	Sensitivity and additional analyses (results)	See 13a–13e	—	See subitems
13a	Robustness sensitivity analyses	Report sensitivity analyses assessing the robustness of the main results to violations of the assumptions	Heterogeneity Q, Egger intercept, and PRESSO showed no pleiotropy evidence that would change the conclusions (Results; Table S6)	Reported
13b	Other sensitivity/additional analysis results	Report results from other sensitivity or additional analyses	coloc.abf across 24 combinations (PPH4/PPH3; Results); SuSiE fine-mapping sensitivity (10/10 converged; conclusions for FBLN5/HEMK1 upgraded to	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
			multi-signal confirmation); comparison against the known-causal gene lists for both diseases (Table S8)	
13c	Assessment of causal direction	Report any assessment of the direction of causality (e.g. bidirectional MR)	Steiger directionality testing passed 32/32 (Results)	Reported
13d	Comparison with non-MR estimates	When relevant, report and compare with estimates from non-MR analyses	The design includes no non-MR causal estimate; direction concordance of candidates was cross-checked across eQTL data sources (Results)	Not applicable
13e	Additional visualization	Consider additional plots (e.g. leave-one-	Fig 5c (abf–SuSiE concordance	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
		out analysis plots)	scatter) and Fig 5d (locuszoom regional LD plots)	
14	Key results	Summarize key results with reference to the study objectives	Discussion (first paragraph) and genetic-layer summary: MR null after correction, 6 nominal candidates, colocalization discriminating signals driven by linked independent variants	Reported
15	Limitations	Discuss limitations in light of the validity of the IV assumptions, other sources of bias, and imprecision; discuss the direction and	Discussion (“MR and population limitations”): power and population-composition explanations for the corrected null, eQTL	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
		magnitude of potential bias and efforts to address it	tissue representativeness, the HEMK1 single-SNP instrument precluding pleiotropy testing, and the reasons the Phe-MR extension was not performed	
16	Interpretation	See 16a–16c	—	See subitems
16a	Meaning	Give a cautious overall interpretation in the context of limitations and in comparison with other studies	Discussion: candidates are leads rather than established causality; positioned as a novel candidate layer relative to the known-causal lists of both diseases; demarcation from population-genetics studies	Reported
16b	Mechanism and	Discuss	Cautious causal	Partially

Item	Topic	Requirement (summary)	Location in this manuscript	Status
	causal language	biological mechanisms that could drive the causal relationship and the plausibility of the gene–environment equivalence assumption; use causal language carefully	language ("candidate leads rather than established causal relationships"; Results); mechanism discussion (Discussion); the gene–environment equivalence assumption not explicitly discussed	reported
16c	Clinical relevance	Discuss clinical/policy relevance of the results and implications for intervention effect sizes	Translational positioning and the statement that "independent validation is required before translation can be considered" (Discussion); Conclusions section	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
17	Generalizability	Discuss generalizability to other populations, other exposure periods/timings, and other exposure levels	Population dimension: extrapolation limits of predominantly European-ancestry samples and the sex composition of the blood cohorts (Discussion); exposure-period and exposure-level dimensions not elaborated	Partially reported
18	Funding	Describe the funding sources of the present study and the role of funders; if applicable, funding sources of the original databases/studies	“Funding”: this study received no dedicated funding (Declarations); data sources fully described in the Data availability section	Reported
19	Data and data sharing	Provide the data used in the	Data availability section	Reported

Item	Topic	Requirement (summary)	Location in this manuscript	Status
		analysis or describe where they can be found, citing sources in the article; provide the statistical code or describe where it is publicly available	(Declarations): sources for GEO/OpenGWAS/eQTLGen 2019-12-11 release/GTEx v8/the druggability list, Tables S1–S28, GitHub code repository (with explicit seeds), and licence restrictions	
20	Conflicts of interest	All authors should declare all potential conflicts of interest	"Competing interests": all authors declare no competing interests (Declarations)	Reported

Gap summary (44 content items)

Status counts: Reported 29; Partially reported 11; Not applicable 4; Not reported 0. (Purely heading-type main items 9/10/11/12/13/16 are not counted separately. Item 9b was verified to be a checklist update omission and item 10c was completed in the main text; both were closed on 2026-10-08.)

Not reported (0 items; the former 2 items were closed on 2026-10-08): 1. ~~**9b Pre-registration**~~ **Closed (verified and removed)**: The Declarations paragraph "Study protocol,

registration, and patient involvement" explicitly states "No separately registered study protocol document was prepared … This study was not registered" and explains that key analytic steps followed a prespecified internal analysis plan—this row reflected a checklist update omission rather than a content gap. 2. ~~**10c Heterogeneity between data sources**~~ **Closed (completed in the main text)**: The "GWAS and QTL resources" paragraph (Methods) now states that between-source-study heterogeneity is as per the original publications and was not re-assessed, and points to the Cochran's Q assessment at the level of this study (Table S6).

Partially reported (11 items, with prioritized suggestions for strengthening): - **4b/10a (participants/numbers at each stage)**: The outcome GWAS sample sizes (appendicular lean mass GCST90000025, low grip strength GCST90007526, walking pace ukb-b-4711, knee OA GCST007090) are not listed in the main text, and no a priori power calculation is described. Suggestion: add the sample sizes and case/control composition of each outcome GWAS to Table S17, and state that no a priori power calculation was performed (precision assessed via 95% CI width). - **10d (two-sample similarity and overlap)**: Suggest quantifying the sample overlap between eQTLGen/GTEx and each outcome GWAS (expected to be minimal), or supporting it with cross-trait LDSC intercepts. - **4d (outcome definitions)**: Suggest adding a brief table of the phenotype definitions/diagnostic criteria for the four outcomes (or explicit pointers to the phenotype sections of the original GWAS publications). - **5/6c (additional assumptions and covariate sets)**: Suggest adding one sentence on the additional assumptions of the sensitivity analyses (InSIDE for MR-Egger, majority-valid for the weighted median) and a note that covariate adjustment under the summary-data design follows the original GWAS. - **6a/11b (scale and uncertainty)**: Suggest confirming whether Tables S6/S17 contain per-combination b, SE, 95% CI, and per-SNP association details, and adding a pointer sentence in the main text; at the main-text level, add 95% CIs for the strongest candidates (e.g. FBLN5–ALM). - **16b (mechanism/causal language)**: Suggest adding a sentence discussing the plausibility of the gene–environment equivalence assumption in the cis-eQTL drug-target context (cis instruments plus a central-dogma directionality argument). - **17 (generalizability)**: Suggest adding one sentence on the exposure-period/exposure-level dimensions (eQTLs reflect adult steady-state expression, so extrapolation to lifetime exposure scales is limited).

Not applicable (4 items; it is recommended to retain the "Not applicable + reason" wording in the submitted version of the checklist): 6d (no individual-level missing data in a summary-data design), 10b (phenotype summary statistics are as per the original GWAS), 11c (results null after correction; absolute-risk translation not meaningful), 13d (no parallel non-MR causal estimate).

Ancillary tracking record (not a checklist item): The page numbers of final-manuscript reference [12] (the STROBE-MR statement) have been corrected to **1614–1621**.

Part 2. Appendix D: known-causal gene control lists for both diseases (13-gene OA version)

D.1 Purpose and inclusion criteria

This appendix lists the known causal/candidate therapeutic-target genes reported by previous single-disease studies on the osteoarthritis (OA) side and the sarcopenia side, serving as the control list for the known-causal overlap of the 10 biomarkers of this study (GADD45A, EGFR, NELL1, STEAP1, FBLN5, RNF14, HEMK1, GDE1, CBR3, BNIP3) (main text; Table S8).

Inclusion criteria: Tier 1/2 targets of each source publication or author-named core targets; this list is used for overlap comparison and is **not an exhaustive enumeration**.

D.2 Source literature

OA side (three publications, 13 genes after deduplication):

1. **Lin et al. 2025 (Current Science)**: Druggable-genome-wide summary-data-based MR (SMR) × 7 OA subtypes and 4 OA-related traits, with stepwise stratified validation by colocalization and two-sample MR/pQTL; **6 tier 3 targets**: MAPK3, CSK, IFNGR2, MVD, FES, ITGA2.
2. **Zou et al. 2024 (Biomolecules; PMC10967895)**: Plasma proteome MR × knee/hip OA (12 FDR-corrected candidate proteins, tier-stratified). **Tier 1**: MAPK3 (knee OA,

OR=0.855), GZMK (knee OA, OR=1.278), ITIH1 (hip OA, OR=0.847); **tier 2**: LGALS3 (knee OA OR=1.120; SMR validation OR=1.638, concordant direction; coloc PPH4=0.828).

3. **Lu et al. (PLoS One; PMC11813149; 1,553 circulating proteins × 5 OA traits)**: After Bonferroni correction, Steiger filtering, colocalization (PPH4>0.8), and phenome scanning, the authors finalized 4 causal proteins: DNAJB12, USP8 (knee OA), IL12B (spinal OA), RGMB (thumb OA).

Version note: LGALS3 is a tier 2 target (knee OA OR=1.120; SMR validation OR=1.638, concordant direction; coloc PPH4=0.828) and was retained upon adjudication (2026-08-29), constituting the difference between the 13-gene and 12-gene versions (the 12-gene version does not include LGALS3).

Sarcopenia side (one publication, 6 genes):

4. **Yin et al. 2024 (J Cachexia Sarcopenia Muscle; doi:10.1002/jcsm.13479)**: Systematic druggable-genome MR (blood/skeletal-muscle eQTL and pQTL × sarcopenia-related traits: appendicular lean mass, grip strength, walking pace); among 17 MR candidates, **6 were confirmed by Bayesian colocalization (PPH4>0.9)**: MAP3K3, MFGE8, COL15A1, HP, HLA-DRA, AURKA.

D.3 Control list (after deduplication: 13 OA + 6 sarcopenia = 19 genes)

Gene	Disease side	Source literature	Target tier/evidence	Overlap with the 10 biomarkers of this study
MAPK3	OA	Lin et al. 2025 (Current Science); Zou et al. 2024 (Biomolecules, PMC10967895)	Lin tier 3; Zou tier 1 (knee OA, plasma protein MR OR=0.855, author-named core target)	None
CSK	OA	Lin et al. 2025 (Current Science)	Tier 3 (SMR + colocalization + two-sample MR/pQTL validation)	None
IFNGR2	OA	Lin et al. 2025 (Current Science)	Tier 3	None
MVD	OA	Lin et al. 2025 (Current Science)	Tier 3	None
FES	OA	Lin et al. 2025 (Current Science)	Tier 3	None
ITGA2	OA	Lin et al. 2025 (Current Science)	Tier 3	None

Gene	Disease side	Source literature	Target tier/evidence	Overlap with the 10 biomarkers of this study
GZMK	OA	Zou et al. 2024 (Biomolecules, PMC10967895)	Tier 1 (knee OA, OR=1.278, author-named core target)	None
ITIH1	OA	Zou et al. 2024 (Biomolecules, PMC10967895)	Tier 1 (hip OA, OR=0.847, author-named core target)	None
LGALS3	OA	Zou et al. 2024 (Biomolecules, PMC10967895)	Tier 2 (knee OA OR=1.120; SMR validation OR=1.638, concordant direction; coloc PPH4=0.828; retained upon adjudication 2026-08-29)	None
DNAJB12	OA	Lu et al. (PLoS One, PMC11813149)	Author-finalized causal protein (knee OA, OR=1.51; PPH4=0.902)	None
USP8	OA	Lu et al. (PLoS One, PMC11813149)	Author-finalized causal protein (knee OA,	None

Gene	Disease side	Source literature	Target tier/evidence	Overlap with the 10 biomarkers of this study
			OR=1.72; PPH4=0.864)	
IL12B	OA	Lu et al. (PLoS One, PMC11813149)	Author-finalized causal protein (spinal OA, OR=0.87; PPH4=0.978)	None
RGMB	OA	Lu et al. (PLoS One, PMC11813149)	Author-finalized causal protein (thumb OA, OR=0.59; PPH4=0.932)	None
MAP3K3	Sarcopenia	Yin et al. 2024 (J Cachexia Sarcopenia Muscle, doi:10.1002/jcsm.13479)	Colocalization-confirmed target (PPH4>0.9; associated with appendicular lean mass, grip strength, and walking pace)	None
MFGE8	Sarcopenia	Yin et al. 2024 (as above)	Colocalization-confirmed target (PPH4>0.9; dual-source validation by muscle eQTL	None

Gene	Disease side	Source literature	Target tier/evidence	Overlap with the 10 biomarkers of this study
			and blood pQTL)	
COL15A1	Sarcopenia	Yin et al. 2024 (as above)	Colocalization-confirmed target (PPH4>0.9; dual-source validation by muscle eQTL and blood pQTL)	None
HP	Sarcopenia	Yin et al. 2024 (as above)	Colocalization-confirmed target (PPH4>0.9; ≥2 sarcopenia-related traits)	None
HLA-DRA	Sarcopenia	Yin et al. 2024 (as above)	Colocalization-confirmed target (PPH4>0.9; ≥2 sarcopenia-related traits)	None
AURKA	Sarcopenia	Yin et al. 2024 (as above)	Colocalization-confirmed target (PPH4>0.9; opposite directions in blood and	None

Gene	Disease side	Source literature	Target tier/evidence	Overlap with the 10 biomarkers of this study
			skeletal muscle, with the authors noting the need for further validation)	

D.4 Control conclusion

The 10 biomarkers of this study (**GADD45A, EGFR, NELL1, STEAP1, FBLN5, RNF14, HEMK1, GDE1, CBR3, BNIP3**) have **no overlap** with either the OA-side 13-gene list (MAPK3, CSK, IFNGR2, MVD, FES, ITGA2, GZMK, LGALS3, ITIH1, DNAJB12, USP8, IL12B, RGMB) or the sarcopenia-side 6-gene list (MAP3K3, MFGE8, COL15A1, HP, HLA-DRA, AURKA) (0/10; overlap analysis record: `results/moduleH/moduleH_known_causal_overlap.csv`; Table S8).

Interpretation (as stated in the main text): The candidate-generation logic of this study is “shared across both diseases plus directionality”, which is orthogonal to the “single-disease causality” logic of single-disease studies; non-overlap is therefore expected. The biomarkers of this study constitute a complementary **novel candidate layer** rather than a rediscovery of existing targets.

Usage limitations: This list was compiled according to the tier 1/2 targets or author-named core targets of each source publication and is not an exhaustive enumeration; other candidates reported by OA MR/TWAS studies (e.g. SMAD3, COL2A1, HHIP, MGP) were not included in the comparison, and the overlap conclusion holds only within the scope of this list.